SET 74 • HSF-IF.C.7.b

BACKWARD 2

Evaluate square roots and cube roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Evaluate $\sqrt{81}$ and $\sqrt[3]{(-27)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q1/5

Solve $x^3 = 125$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q2/5

What is $\sqrt[3]{8} + \sqrt{4}$?

A. 6

B. 12

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{0.25} = 0.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q4/5

True or false?

$\sqrt[3]{(-8)}$ is not a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • HSF-IF.C.7.b

FORWARD 1

Solve radical equations and check the solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x + 4)} = 3$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q1/5

Solve $\sqrt[3]{(2x)} = -2$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q2/5

Solve $\sqrt{(x + 2)} = x$.

A. $x = -1$ only

B. $x = 2$ or $x = -1$

C. No solution

D. $x = 2$ only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 74 • FORWARD 1 • Q3/5

True or false?

$\sqrt{(x - 3)} = -2$ has no real solution.

A. True

B. False

True or false?

Cubing both sides of $\sqrt[3]{x} = 4$ gives $x = 12$.

A. True

B. False

SET 74 • HSF-IF.C.7.b

FORWARD 2

Find inverses of square root and cube root functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

$f(x) = \sqrt{x}$ for $x \geq 0$. Find $f^{-1}(5)$.

$f^{-1}(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q1/5

$g(x) = \sqrt[3]{x}$. Find $g^{-1}(-2)$.

$g^{-1}(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q2/5

Which is the inverse of $f(x) = \sqrt{(x - 1)}$, for $x \geq 1$?

A. $f^{-1}(x) = \sqrt{(x + 1)}$

B. $f^{-1}(x) = x^2 + 1$, for $x \geq 0$

C. $f^{-1}(x) = x^2 - 1$

D. $f^{-1}(x) = (x + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S74-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 74 • FORWARD 2 • Q3/5

True or false?

The graph of $y = \sqrt{x}$ $(x \geq 0)$ is the reflection of the graph of $y = x^2$ $(x \geq 0)$ across the line $y = x$.

A. True

B. False

True or false?

$g(x) = \sqrt[3]{x}$ has no inverse function.

A. True

B. False

SET 75 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph piecewise-defined and step functions

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate piecewise-defined functions	**HSF-IF.A.2**
BACKWARD 2	Order numbers and describe intervals on a number line	**6.NS.C.7.a**
FORWARD 1	Graph the probability distribution of a discrete random variable —..	**HSS-MD.A.1**
FORWARD 2	Read inverse values of piecewise and step functions	**HSF-BF.B.4.c**

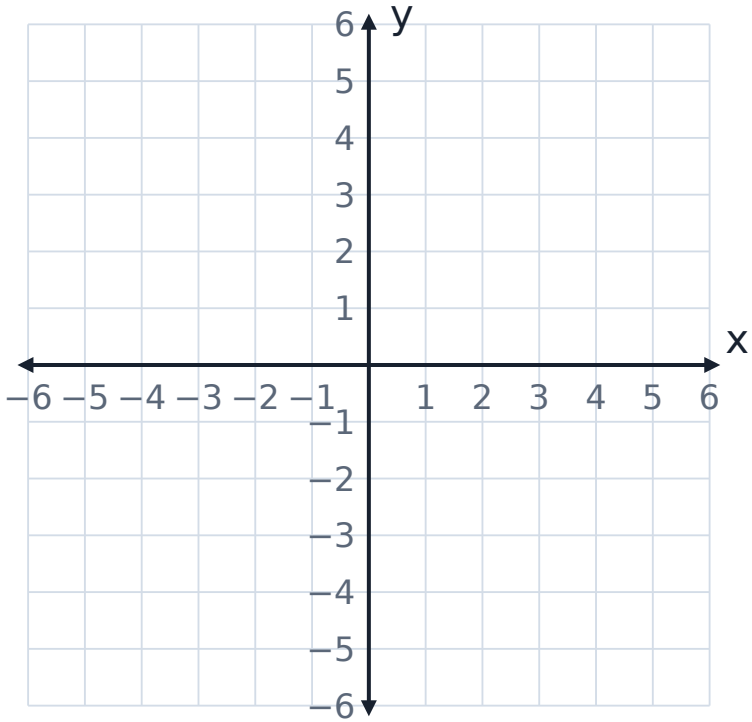
SET 75 • HSF-IF.C.7.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

$f(x) = x + 1$ for $x < 2$, and
$f(x) = 5 - x$ for $x \geq 2$.

Graph f. Then find f(0), f(2), and f(4).



Answer: _____

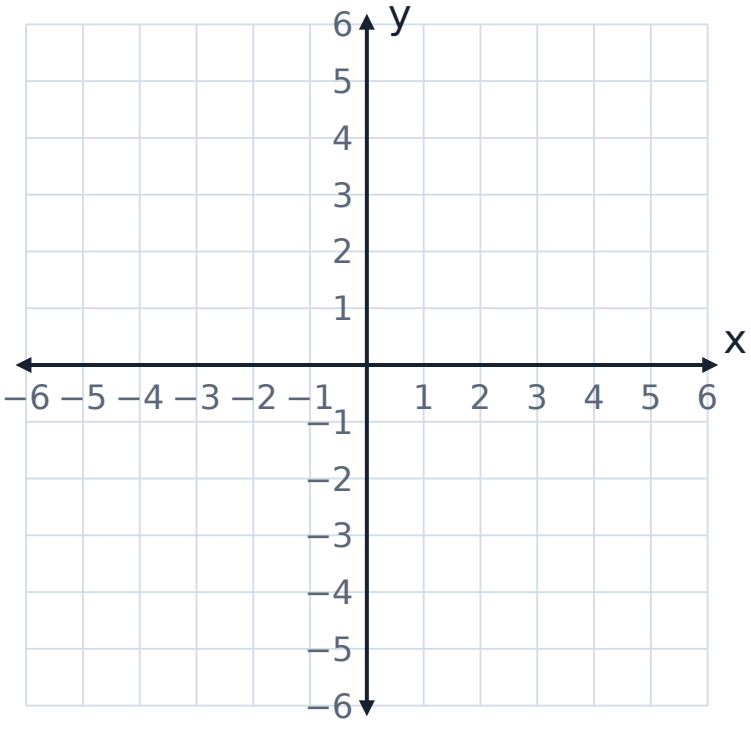
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q1/5

$g(x) = -1$ for $x < 0$, $g(x) = 2$ for $0 \leq x < 3$, and $g(x) = x - 1$ for $x \geq 3$.

Graph g. Use open and closed dots at the breaks.

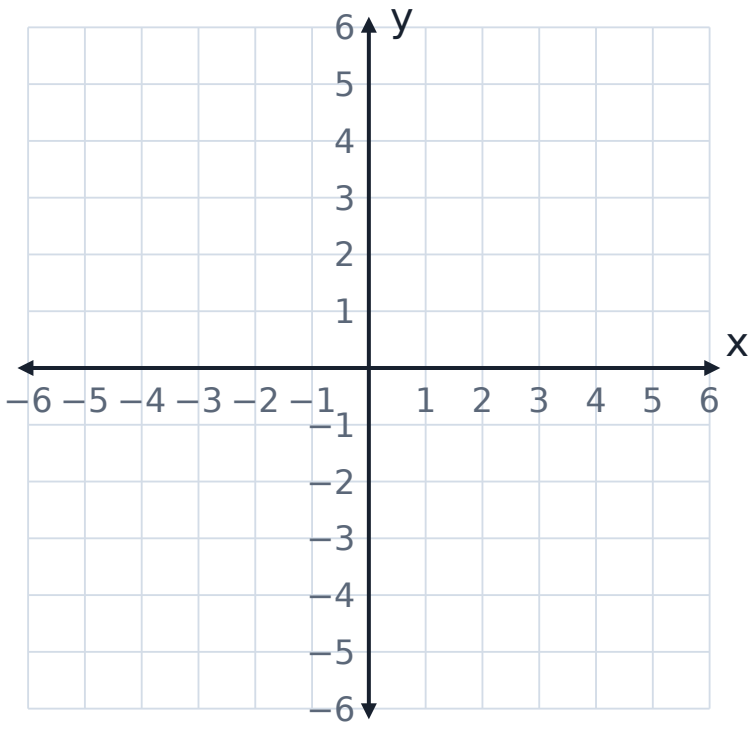


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q2/5

Graph the greatest integer function $y = \lfloor x \rfloor$ (the greatest integer less than or equal to x) for $-2 \leq x < 3$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q3/5

A parking garage charges $3 for each hour or part of an hour.

What is the charge for 2.25 hours?

A. $6

B. $6.75

C. $7.50

D. $9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 75 • MAIN • Q4/5

True or false?

For the greatest integer function, $\lfloor 2.7 \rfloor = 3$.

A. True

B. False

SET 75 • HSF-IF.C.7.b

BACKWARD 1

Evaluate piecewise-defined functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(x) = 2x - 1$ for $x < 3$, and $f(x) = 8$ for $x \geq 3$.

Find $f(3)$ and $f(1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q1/5

$g(x) = |x|$ for $x \leq 0$, and $g(x) = x^2$ for $x > 0$.

Find $g(-4)$ and $g(4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q2/5

$h(x) = 3$ for $x < 1$, and $h(x) = 2x$ for $x \geq 1$. What is $h(1)$?

A. 2

B. 5

C. 3

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 75 • BACKWARD 1 • Q3/5

True or false?

$p(x) = 5$ for $x \leq 2$, and $p(x) = x$ for $x > 2$. Then $p(2) = 5$.

A. True

B. False

True or false?

$\lfloor -1.5 \rfloor = -1$

A. True

B. False

SET 75 • HSF-IF.C.7.b

BACKWARD 2

Order numbers and describe intervals on a number line • Grade 6 • The Number System • 6.NS.C.7.a

Is −3 to the left or to the right of −1 on a number line? Which number is greater?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q1/5

Write an inequality for "x is at least 0 and less than 3".

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q2/5

Which statement is true?

A. $-4 > -2$

B. $-4 < -2$

C. $0 < -1$

D. $-2 < -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q3/5

True or false?

-5 is less than -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q4/5

True or false?

On a number line, 2 is to the left of −3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 75 • BACKWARD 2 • Q5/5

FORWARD 1

Graph the probability distribution of a discrete random variable • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: A probability distribution assigns a value to each separate outcome, like a function defined piece by piece, and its bar graph is read the same way.

A fair coin is flipped twice, and X is the number of heads. Give P(X = 0), P(X = 1), and P(X = 2).

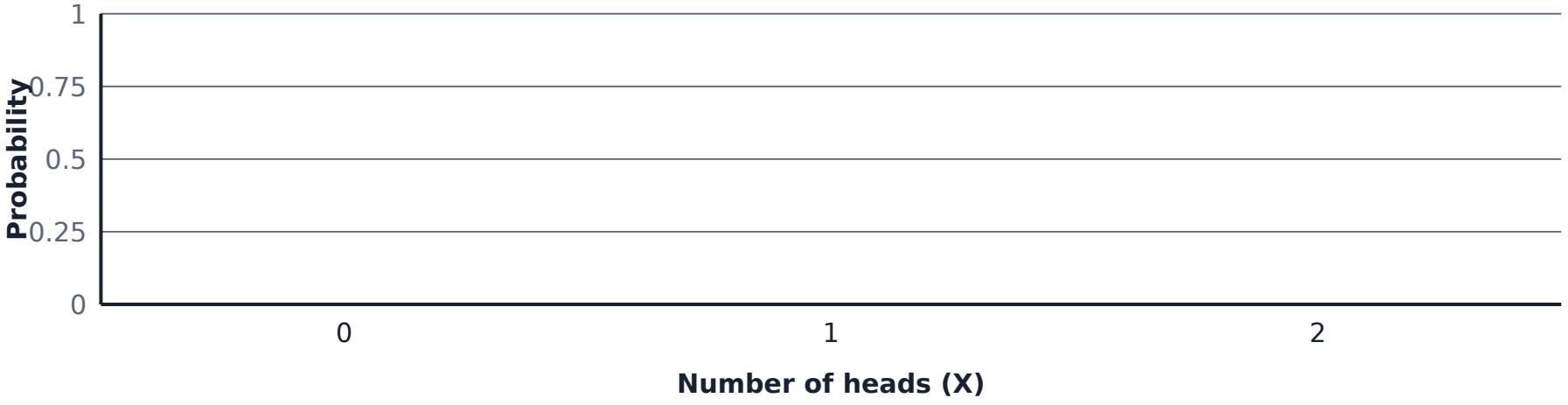
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q1/5 • NEAREST RELATED

A fair coin is flipped twice, and X is the number of heads. Graph the probability distribution of X as a bar graph.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions HSS-MD.A.1 • Set 75 • FORWARD 1 • Q2/5 • NEAREST RELATED

X is the result of one roll of a fair number cube numbered 1 to 6. What is $P(X \geq 5)$?

A. $\frac{5}{6}$

B. $\frac{1}{6}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

The probabilities of a probability distribution add to 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

For X = the number of heads in 2 fair coin flips, $P(X = 1) = \frac{1}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 75 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 75 • HSF-IF.C.7.b

FORWARD 2

Read inverse values of piecewise and step functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

$f(x) = 2x$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$. This f is one-to-one. Find $f^{-1}(9)$ and $f^{-1}(-4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q1/5

$g(x) = x + 1$ for $x < 2$, and $g(x) = 2x - 1$ for $x \geq 2$. This g is one-to-one. Find $g^{-1}(7)$.

$g^{-1}(7) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q2/5

Why does the greatest integer function $y = \lfloor x \rfloor$ have no inverse function?

A. Many inputs share each output, such as $\lfloor 2.1 \rfloor = \lfloor 2.9 \rfloor = 2$.

B. It is a linear function.

C. It is not defined for negative x.

D. Its outputs are not integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q3/5

True or false?

A piecewise function whose pieces are increasing and whose output ranges do not overlap can have an inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q4/5

True or false?

If f is one-to-one and $f(3) = 5$, then $f^{-1}(3) = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 75 • FORWARD 2 • Q5/5

SET 44 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph exponential functions and show intercepts and end behavior

MAIN	Algebra I main question	**HSF-IF.C.7.e**
BACKWARD 1	Evaluate integer exponents	**8.EE.A.1**
BACKWARD 2	Evaluate exponential functions	**HSF-IF.A.2**
FORWARD 1	Solve exponential equations with logarithms	**HSF-LE.A.4**
FORWARD 2	Understand logarithms as inverses of exponential functions	**HSF-BF.B.5**

SET 44 • HSF-IF.C.7.e

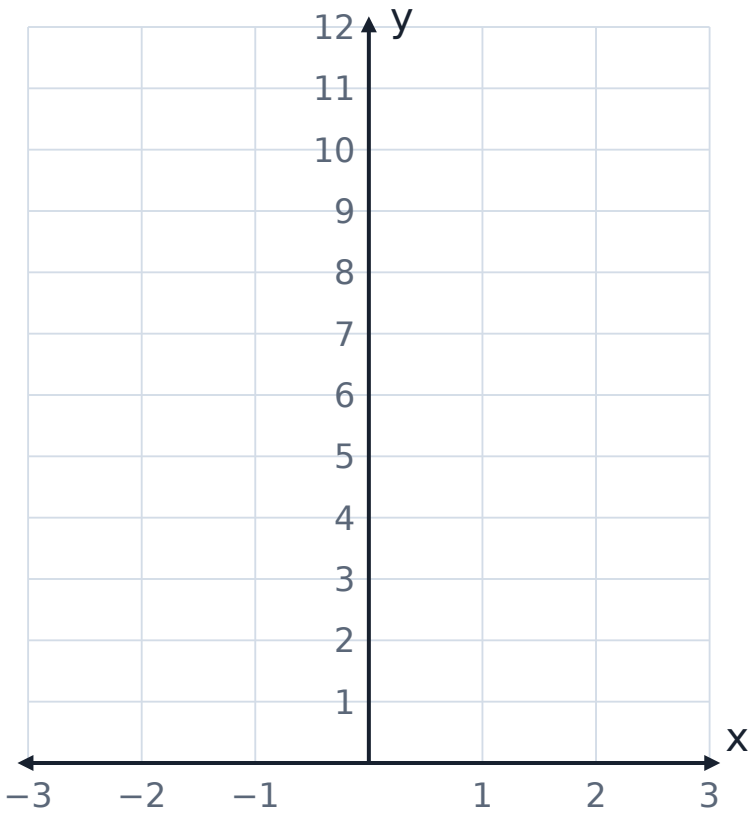
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph $f(x) = 3 \cdot 2^x$ for $-2 \leq x \leq 2$.

Give its y-intercept and its horizontal asymptote.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q1/5

$g(x) = 50(0.8)^x$.

Is this growth or decay? Give its y-intercept, and say what happens to g(x) as x gets large.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q2/5

$h(x) = 2^x - 4$.

Find the y-intercept, the x-intercept, and the horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q3/5

Which function's graph rises from left to right and passes through (0, 5)?

A. $y = 1.2(5)^x$

B. $y = 0.8(5)^x$

C. $y = 5(0.8)^x$

D. $y = 5(1.2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q4/5

True or false?

The graph of $y = 4(0.5)^x$ crosses the x-axis.

A. True

B. False

SET 44 • HSF-IF.C.7.e

BACKWARD 1

Evaluate integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Evaluate 2^{-2}.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

Evaluate $3 \cdot 2^{-1}$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

Which number equals $(0.5)^{-1}$?

A. -0.5

B. 0.5

C. 2

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

True or false?

$5^0 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

True or false?

$4^{-1} = \frac{1}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSF-IF.C.7.e

BACKWARD 2

Evaluate exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(x) = 50(0.8)^x$.

Find f(2).

f(2) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q1/5

$g(x) = 2^x - 4$.

Find $g(3)$.

$g(3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q2/5

$h(x) = 3 \cdot 2^x$.

Find $h(-1)$.

A. -1.5

B. 6

C. -6

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q3/5

True or false?

If $f(x) = 10^x$, then $f(3) = 1{,}000$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q4/5

True or false?

If $f(x) = 3^x$, then $f(2) = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSF-IF.C.7.e

FORWARD 1

Solve exponential equations with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $3 \cdot 2^x = 48$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q1/5

Solve $50(0.8)^x = 20$. Round to the nearest tenth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q2/5

Solve $2^x = 12$. (log means base 10.)

A. $x = \log 6$

B. $x = \log 12 \div \log 2 \approx 3.58$

C. $x = 3.5$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q3/5

True or false?

log 1 = 0 (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q4/5

True or false?

log 100 = 2 (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSF-IF.C.7.e

FORWARD 2

Understand logarithms as inverses of exponential functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

Write the inverse of $f(x) = 2^x$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q1/5

Explain why $\log_3 81 = 4$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q2/5

The graph of $y = \log_2 x$ is the reflection of the graph of $y = 2^x$ across which line?

A. $x = 0$

B. $y = 0$

C. $y = x$

D. $y = -x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q3/5

True or false?

The domain of $y = \log x$ is $x > 0$. (log means base 10.)

A. True

B. False

True or false?

$\log_2(2^5) = 2$

A. True

B. False

SET 45 • HSF-IF.C.8.a • INTERPRETING FUNCTIONS

Factor and complete the square to show zeros, extreme values, and symmetry, and interpret them

MAIN	Algebra I main question	**HSF-IF.C.8.a**
BACKWARD 1	Complete the square	**HSA-SSE.B.3.b**
BACKWARD 2	Use the greatest common factor	**6.NS.B.4**
FORWARD 1	Prove polynomial identities	**HSA-APR.C.4**
FORWARD 2	Complete the square to find the center of a hyperbola	**HSG-GPE.A.3**

SET 45 • HSF-IF.C.8.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a

$f(x) = x^2 - 8x + 12$.

Write f in factored form and in vertex form. Give the zeros, the vertex, and the axis of symmetry.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q1/5

A company's profit is $P(x) = -2x^2 + 120x - 1{,}000$ dollars when it charges x dollars.

Complete the square to find the price that gives the most profit, and the greatest profit.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q2/5

A rock's height is $h(t) = -16t^2 + 32t + 48$ feet after t seconds.

Factor to find when it hits the ground.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q3/5

Which form of $f(x) = x^2 + 6x + 5$ shows the vertex directly?

A. $(x + 1)(x + 5)$

B. $x^2 + 6x + 5$

C. $(x + 3)^2 - 4$

D. $x(x + 6) + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q4/5

True or false?

The axis of symmetry of $y = (x - 1)(x - 7)$ is $x = 4$.

A. True

B. False

SET 45 • HSF-IF.C.8.a

BACKWARD 1

Complete the square • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Complete the square.

$x^2 - 8x + 12$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q1/5

Complete the square.

$x^2 + 3x$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q2/5

Which expression equals $-(x^2 - 6x)$?

A. $-(x - 3)^2 - 9$

B. $-(x - 3)^2 + 9$

C. $(x - 3)^2 - 9$

D. $-(x + 3)^2 + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q3/5

True or false?

$$x^2 + 2x + 10 = (x + 1)^2 + 9$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q4/5

True or false?

$(x - 5)^2 - 3$ has its minimum at $x = -5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSF-IF.C.8.a

BACKWARD 2

Use the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write 16 + 40 as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q1/5

Find the greatest common factor of 32 and 48.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q2/5

Which expression equals 36 + 27 and uses the greatest common factor?

A. 9(4 + 3)

B. 3(12 + 6)

C. 6(6 + 27)

D. 9(4 + 27)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q3/5

True or false?

The greatest common factor of 12 and 18 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q4/5

True or false?

$20 + 50 = 5(4 + 5)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSF-IF.C.8.a

FORWARD 1

Prove polynomial identities • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Show that $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ by expanding both sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q1/5

Use $x = 2$ and $y = 1$ in $x^2 - y^2$, $2xy$, and $x^2 + y^2$ to make a Pythagorean triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q2/5

Which equation is an identity (true for all numbers a and b)?

A. $a^3 + b^3 = (a + b)^3$

B. $a^3 - b^3 = (a - b)^3$

C. $a^2 + b^2 = (a + b)^2$

D. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q3/5

True or false?

$$(a + b)^2 - (a - b)^2 = 4ab$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q4/5

True or false?

$(x + 1)^3 = x^3 + 1$

A. True

B. False

SET 45 • HSF-IF.C.8.a

FORWARD 2

Complete the square to find the center of a hyperbola • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

Complete the square to find the center of the hyperbola $x^2 - 4y^2 - 6x - 8y - 11 = 0$.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q1/5

The hyperbola $x^2 \div 16 - y^2 \div 9 = 1$ has foci $(5, 0)$ and $(-5, 0)$.

Show that $(5, 2.25)$ is on the hyperbola and that the absolute difference of its distances to the foci is 8.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q2/5

Which equation is a hyperbola?

A. $y = x^2 \div 4$

B. $x^2 \div 4 + y^2 \div 9 = 1$

C. $x^2 \div 4 - y^2 \div 9 = 1$

D. $x^2 + y^2 = 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q3/5

True or false?

For a hyperbola, the absolute value of the difference of the distances from any of its points to the two foci is constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q4/5

True or false?

The hyperbola $x^2 - y^2 = 1$ passes through $(0, 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q5/5

SET 46 • HSF-IF.C.8.b • INTERPRETING FUNCTIONS

Use the properties of exponents to interpret growth and decay rates

MAIN	Algebra I main question	**HSF-IF.C.8.b**
BACKWARD 1	Read growth and decay from an exponential function	**HSF-IF.C.7.e**
BACKWARD 2	Solve percent increase and decrease problems	**7.RP.A.3**
FORWARD 1	Sum a geometric series in a savings context	**HSA-SSE.B.4**
FORWARD 2	Find times in exponential models with logarithms	**HSF-LE.A.4**

SET 46 • HSF-IF.C.8.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

A savings balance is $y = 800(1.035)^t$ dollars after t years.

Is this growth or decay, and what is the percent rate of change per year?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q1/5

A loan balance grows by 1.5% each month:

$B = 2{,}000(1.015)^{12t}$, where t is in years.

Rewrite it to find the yearly growth factor (to four decimal places) and the yearly percent rate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q2/5

The value of a machine is $V = 45{,}000(0.88)^t$ dollars after t years.

What is the yearly percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q3/5

A forest's area is $A = 600(1.3)^{t/20}$ hectares after t years.

By what percent does A grow every 20 years?

A. 1.3%

B. 13%

C. 3%

D. 30%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q4/5

True or false?

$y = 100(0.5)^{t/8}$ halves every 8 units of t.

A. True

B. False

SET 46 • HSF-IF.C.8.b

BACKWARD 1

Read growth and decay from an exponential function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Is $y = 30(1.5)^x$ increasing or decreasing? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q1/5

What is the y-intercept of $y = 7(0.9)^x$?

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q2/5

Which function is decreasing?

A. $y = 2(1.6)^x$

B. $y = 0.6(1.2)^x$

C. $y = 2(0.6)^x$

D. $y = 0.6(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q3/5

True or false?

The graph of $y = 3^x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q4/5

True or false?

$y = 5(1.1)^x$ has y-intercept 1.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSF-IF.C.8.b

BACKWARD 2

Solve percent increase and decrease problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A $40 shirt is marked up 15%.

What is the new price?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q1/5

A town grows from 2,000 to 2,300 people.

What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q2/5

A price drops from $80 to $60.

What is the percent decrease?

A. 25%

B. 20%

C. 33%

D. 75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q3/5

True or false?

Multiplying by 1.06 increases a number by 6%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q4/5

True or false?

Multiplying by 0.85 decreases a number by 85%.

A. True

B. False

SET 46 • HSF-IF.C.8.b

FORWARD 1

Sum a geometric series in a savings context • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

You save $100 the first month and 2% more each month after that.

Write the total for 12 months as a sum formula, and find it to the nearest dollar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q1/5

Find $4 + 4(1.5) + 4(1.5)^2 + 4(1.5)^3$.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q2/5

Which expression is the sum of the first n terms of $a + ar + ar^2 + \ldots$ $(r \neq 1)$?

A. $a \div (1 - r)$

B. $a(1 - r) \div (1 - r^n)$

C. $a(1 - r^n) \div (1 - r)$

D. nar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q3/5

True or false?

$$1 + 0.5 + 0.25 + 0.125 = 1.875$$

A. True

B. False

True or false?

The sum of 10 terms of $3 + 3 + 3 + \ldots$ is 3^{10}.

A. True

B. False

SET 46 • HSF-IF.C.8.b

FORWARD 2

Find times in exponential models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$y = 100(0.5)^{t/8}$.

Find t when y = 10. Round to the nearest tenth.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q1/5

Find the continuous rate r for which $e^{10r} = 2$. Round to the nearest thousandth.

$r \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q2/5

To the nearest tenth, how long does $y = (1.05)^t$ take to double?

A. 20

B. 7.2

C. 10.5

D. 14.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q3/5

True or false?

If $y = 2^{t/3}$, then y doubles every 3 units of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q4/5

True or false?

$\log(2^{10}) = 20$ (log means base 10.)

A. True

B. False

SET 47 • HSF-IF.C.9 • INTERPRETING FUNCTIONS

Compare properties of two functions represented in different ways

MAIN	Algebra I main question	**HSF-IF.C.9**
BACKWARD 1	Interpret key features in context	**HSF-IF.B.4**
BACKWARD 2	Compute average rates of change	**HSF-IF.B.6**
FORWARD 1	Compare periodic functions given in different forms	**HSF-TF.B.5**
FORWARD 2	Compare normal distributions by mean and standard deviation —…	**HSS-ID.A.4**

SET 47 • HSF-IF.C.9

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9

f(x) = 3x + 2. The table shows g.

Which function has the greater value at x = 3? Which has the greater average rate of change from x = 0 to x = 3?

x	0	1	2	3
g(x)	1	4	9	16

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q1/5

Function A is $y = -2(x - 3)^2 + 10$. Function B is a parabola that opens down with vertex (1, 12).

Which has the greater maximum value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q2/5

Function f has y-intercept 4 and increases by 5 for each 1-unit increase in x. $g(x) = 5x - 1$.

Compare their slopes and y-intercepts.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q3/5

$P(x) = 1.08^x$ grows 8% per year. The table shows Q.

Which grows faster?

x	0	1	2
Q(x)	1	1.1	1.21

A. P: its graph is a line.

B. Q: it grows 10% per year.

C. P: it grows 8% per year and Q grows 1%.

D. They grow at the same rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q4/5

True or false?

$f(x) = x^2$, and g is a parabola opening up with vertex $(0, -4)$. The two functions have the same minimum value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q5/5

SET 47 • HSF-IF.C.9

BACKWARD 1

Interpret key features in context • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

T(h) is the temperature h hours after midnight, and its minimum is T(6) = 4.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q1/5

A graph of savings S(m) is horizontal from m = 3 to m = 5.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q2/5

A function's graph is an unbroken curve. It is positive for $0 < x < 8$ and negative for $x > 8$.

What is true at $x = 8$?

A. The output is greatest.

B. The output is 8.

C. The output is 0.

D. The function is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q3/5

True or false?

A y-intercept gives the starting value of a quantity that changes over time.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q4/5

True or false?

An x-intercept tells the starting value of a quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSF-IF.C.9

BACKWARD 2

Compute average rates of change • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = 2^x$.

Find the average rate of change from $x = 1$ to $x = 3$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q1/5

A savings account had $100 at week 0 and $180 at week 4.

What is the average rate of change?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q2/5

$g(x) = x^2 - 1$.

What is the average rate of change from $x = -1$ to $x = 2$?

A. 1

B. -1

C. 3

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q3/5

True or false?

The average rate of change of $y = 5$ is 0 on every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q4/5

True or false?

The average rate of change of $y = x^2$ from $x = 0$ to $x = 2$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSF-IF.C.9

FORWARD 1

Compare periodic functions given in different forms • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

$f(x) = 3\sin(2x)$. Function g has amplitude 4 and period π.

Which has the greater amplitude? Do they have the same period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q1/5

Tide A rises to 12 ft and falls to 2 ft. Tide B is $h(t) = 6 + 4\sin t$.

Which has the greater amplitude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q2/5

What is the period of $y = \sin(4x)$?

A. $\pi \div 2$

B. $\pi \div 4$

C. 2π

D. 4π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q3/5

True or false?

$y = 2\cos x$ and $y = 2\sin x$ have the same amplitude and the same period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q4/5

True or false?

$y = 5\sin x$ has a maximum value of 10.

A. True

B. False

SET 47 • HSF-IF.C.9

FORWARD 2

Compare normal distributions by mean and standard deviation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Comparing two functions by their key features leads to comparing two normal curves by their center and spread.

Test A scores are normal with mean 70 and standard deviation 5. Test B scores are normal with mean 70 and standard deviation 10.

Which test's scores are more spread out?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 47 • FORWARD 2 • Q1/5 • NEAREST RELATED

Heights are normal with mean 64 in and standard deviation 3 in.

About what percent are between 61 in and 67 in?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q2/5 • NEAREST RELATED

Scores are normal with mean 100 and standard deviation 15.

About what percent of scores are above 130?

A. About 16%

B. About 34%

C. About 2.5%

D. About 5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A normal distribution is symmetric about its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 47 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

In a normal distribution, about 95% of the data lie within 1 standard deviation of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 47 • FORWARD 2 • Q5/5 • NEAREST RELATED

Write an explicit expression or a recursive process for a function from a context

MAIN	Algebra I main question	**HSF-BF.A.1.a**
BACKWARD 1	Evaluate a sequence from a recursive or explicit rule	**HSF-IF.A.3**
BACKWARD 2	Write equations in two or more variables	**HSA-CED.A.2**
FORWARD 1	Build a composite function from a context	**HSF-BF.A.1.c**
FORWARD 2	Find and interpret the inverse of a model	**HSF-BF.B.4.b**

SET 48 • HSF-BF.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

A phone plan has a $60 activation fee and costs $35 per month.

Write an explicit expression for the total cost C(n) after n months, and a recursive rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q1/5

A tree is 4 ft tall and grows 1.5 ft per year.

Write a function for its height h(t) after t years.

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q2/5

A ball is dropped from 10 m. Each bounce reaches 60% of the height of the bounce before.

Write a function for the height h(n) of the nth bounce.

h(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q3/5

A table shows n = 1, 2, 3, 4 and a(n) = 7, 11, 15, 19.

Which recursive process fits?

A. $a(1) = 4,\ a(n) = a(n - 1) + 7$

B. $a(1) = 7,\ a(n) = 4a(n - 1)$

C. $a(1) = 7,\ a(n) = a(n - 1) + 4$

D. $a(n) = 7n + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q4/5

True or false?

The steps "start at 500 and multiply by 1.03 each year" give f(t) = 500 + 1.03t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q5/5

SET 48 • HSF-BF.A.1.a

BACKWARD 1

Evaluate a sequence from a recursive or explicit rule • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

$f(1) = 2$ and $f(n) = 3f(n - 1)$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q1/5

$f(n) = 10 - 2n$ for $n \geq 1$.

List the first four terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q2/5

$f(1) = 5$ and $f(n) = f(n - 1) - 3$.

What is $f(3)$?

A. -1

B. 11

C. 2

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q3/5

True or false?

The terms f(1), f(2), f(3) of a sequence are its outputs for the inputs 1, 2, 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q4/5

True or false?

$f(n) = n^2$ gives the terms 1, 2, 3, 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSF-BF.A.1.a

BACKWARD 2

Write equations in two or more variables • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

A car rental costs $45 per day plus $0.20 per mile.

Write an equation for the cost C for d days and m miles.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q1/5

Write an equation for the perimeter P of a rectangle with length l and width w.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q2/5

A $900 laptop loses $150 of value each year.

Which equation gives its value V after t years?

A. $V = 900 - 150t$

B. $V = 150 - 900t$

C. $V = 900(150)^t$

D. $V = 900 + 150t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q3/5

True or false?

$A = s^2$ gives the area of a square with side s.

A. True

B. False

True or false?

y = 12x gives the number of feet in x inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSF-BF.A.1.a

FORWARD 1

Build a composite function from a context • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

An oil spill's radius is $r(t) = 2t$ meters after t hours, and the area of a circle is $A(r) = \pi r^2$.

Write $A(r(t))$ and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q1/5

A store takes 20% off and then $5 off: $d(x) = 0.8x$ and $c(x) = x - 5$.

Find $c(d(50))$.

$c(d(50)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q2/5

With $d(x) = 0.8x$ and $c(x) = x - 5$, which expression is $d(c(x))$?

A. $0.8x - 4$

B. $0.8x + 4$

C. $0.8x - 5$

D. $x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q3/5

True or false?

A store takes 20% off and then $5 off: $d(x) = 0.8x$ and $c(x) = x - 5$. For these functions, $c(d(50))$ and $d(c(50))$ are different.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q4/5

True or false?

If $f(x) = 3x$ and $g(x) = x + 2$, then $f(g(x)) = 3x + 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSF-BF.A.1.a

FORWARD 2

Find and interpret the inverse of a model • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

$C(n) = 60 + 35n$ is the total cost after n months.

Find the inverse and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q1/5

Show that f(x) = 5x − 2 and g(x) = (x + 2) ÷ 5 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q2/5

$F = 1.8C + 32$ converts Celsius to Fahrenheit.

Which formula is its inverse?

A. $C = F \div 1.8 - 32$

B. $C = 1.8F + 32$

C. $C = (F + 32) \div 1.8$

D. $C = (F - 32) \div 1.8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q3/5

True or false?

If f is invertible and $f(2) = 9$, then $f^{-1}(9) = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q4/5

True or false?

The inverse of $f(x) = 2x$ is $f^{-1}(x) = -2x$.

A. True

B. False

Build quadratic functions that model situations

MAIN	Algebra I main question	**HSF-BF.A.1.a**
BACKWARD 1	Write equations in two variables for areas and perimeters	**HSA-CED.A.2**
BACKWARD 2	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**
FORWARD 1	Use quadratic models to make the best design choice	**HSG-MG.A.3**
FORWARD 2	Compose functions to build quadratic models	**HSF-BF.A.1.c**

SET 80 • HSF-BF.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

A farmer has 100 m of fence for three sides of a rectangular pen; a barn wall forms the fourth side. Each side perpendicular to the wall is x m long.

Write the area A(x) of the pen.

A(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q1/5

A theater sells 300 tickets at $20 each. For each $1 increase in price, it sells 10 fewer tickets.

Write the revenue R(n) after n price increases of $1.

R(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q2/5

A stone is thrown upward at 20 m/s from the top of a 5-m wall. Using $h(t) = -4.9t^2 + v_0t + h_0$, where v_0 is the starting upward speed and h_0 is the starting height, write the stone's height h(t).

$h(t) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q3/5

A table has x = 0, 1, 2, 3 and y = 3, 5, 11, 21.

Which function fits the table?

A. $y = 3(2)^x$

B. $y = 2x + 3$

C. $y = 2x^2 + 3$

D. $y = x^2 + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q4/5

True or false?

If each of n people shakes hands once with every other person, the number of handshakes is $h(n) = n(n - 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 80 • MAIN • Q5/5

SET 80 • HSF-BF.A.1.a

BACKWARD 1

Write equations in two variables for areas and perimeters • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

A rectangle has length 2w and width w. Write an equation for its perimeter P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q1/5

A triangle has base b and height 2b. Write an equation for its area A.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q2/5

A rectangle has perimeter 40 and width w. Which equation gives its area A?

A. $A = w^2 + 20$

B. $A = w(40 - w)$

C. $A = 20w$

D. $A = w(20 - w)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q3/5

True or false?

The area of a circle as a function of its radius is $A = \pi r^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q4/5

True or false?

If a rectangle's length is its width w plus 2, then its area is w + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • HSF-BF.A.1.a

BACKWARD 2

Interpret the parts of an expression in context • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

In $R = (20 + n)(300 - 10n)$, what does the factor $300 - 10n$ represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q1/5

In $A = x(100 - 2x)$ for a pen against a wall, what does $100 - 2x$ represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q2/5

In $h = -4.9t^2 + 20t + 5$, what does 5 represent?

A. The starting upward speed

B. The starting height

C. The effect of gravity

D. The time in the air

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q3/5

True or false?

In $-4.9t^2 + 20t + 5$, the coefficient of t is 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q4/5

True or false?

The constant term of $2w^2 + 3w$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 80 • BACKWARD 2 • Q5/5

SET 80 • HSF-BF.A.1.a

FORWARD 1

Use quadratic models to make the best design choice • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

For a pen with area $A(x) = x(100 - 2x)$, which x gives the greatest area, and what is that area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q1/5

For $R(n) = (20 + n)(300 - 10n)$, which number of \$1 increases gives the most revenue? What are the ticket price and the revenue?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q2/5

A 24-m rope is laid out as a rectangle. Which dimensions give the greatest area?

A. 2 m by 10 m

B. 12 m by 12 m

C. 6 m by 6 m

D. 4 m by 8 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q3/5

True or false?

Of all rectangles with perimeter 40, the square has the greatest area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q4/5

True or false?

For $A(x) = x(100 - 2x)$, the area is greatest when $x = 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 80 • FORWARD 1 • Q5/5

SET 80 • HSF-BF.A.1.a

FORWARD 2

Compose functions to build quadratic models • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

Ticket demand is $q(p) = 500 - 10p$ tickets at a price of p dollars, and the price after n increases is $p(n) = 20 + n$.

Write $q(p(n))$ and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q1/5

A square's area is $A(s) = s^2$, and its side grows with time: $s(t) = 2 + 0.5t$.

Write $A(s(t))$ in standard form.

$A(s(t)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q2/5

$f(x) = x^2$ and $g(x) = x + 3$. Which is $f(g(x))$ in standard form?

A. $x^2 + 3x$

B. $x^2 + 9$

C. $x^2 + 3$

D. $x^2 + 6x + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S80-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q3/5

True or false?

If $f(x) = x^2$ and $g(x) = 2x$, then $g(f(3)) = 18$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q4/5

True or false?

If $f(x) = x^2$ and $g(x) = 2x$, then $f(g(3)) = 18$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 80 • FORWARD 2 • Q5/5

Combine standard function types using arithmetic operations

MAIN	Algebra I main question	**HSF-BF.A.1.b**
BACKWARD 1	Add, subtract, and multiply polynomials	**HSA-APR.A.1**
BACKWARD 2	Evaluate and interpret functions	**HSF-IF.A.2**
FORWARD 1	Rewrite rational expressions such as average cost	**HSA-APR.D.6**
FORWARD 2	Graph and interpret an average-cost rational function	**HSF-IF.C.7.d**

SET 49 • HSF-BF.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

Revenue is R(x) = 25x and cost is C(x) = 400 + 9x dollars for x items.

Write the profit function P(x) in simplest form and find P(50).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q1/5

A ball's height is $h(t) = -4.9t^2 + 30t + 2$ meters after t seconds.

Name the three simple functions added together, and what each one describes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q2/5

$f(x) = x^2 + 1$ and $g(x) = 2x - 3$.

Find $(f - g)(x)$ in simplest form, and find $(f \cdot g)(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q3/5

The temperature of tea is $T(t) = 20 + 70(0.9)^t$ °C after t minutes.

Which description fits this combination of functions?

A. A linear function with slope 0.9

B. An exponential function that grows 90% per minute

C. A constant room temperature of 20 °C plus a difference of $70(0.9)^t$ that decays

D. A constant temperature of 90 °C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q4/5

True or false?

If $f(x) = 3x$ and $g(x) = x^2$, then $(f + g)(2) = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q5/5

SET 49 • HSF-BF.A.1.b

BACKWARD 1

Add, subtract, and multiply polynomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Subtract.

$25x - (400 + 9x)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q1/5

Add.

$(x^2 + 1) + (2x - 3)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q2/5

Multiply.

$(x^2 + 1)(2x - 3)$

A. $2x^3 + 2x - 3$

B. $2x^2 + 2x - 3$

C. $2x^3 - 3x^2 - 3$

D. $2x^3 - 3x^2 + 2x - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-B1-Q3**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q3/5

True or false?

The product of two polynomials is a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q4/5

True or false?

$(x + 2) - (x - 2) = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSF-BF.A.1.b

BACKWARD 2

Evaluate and interpret functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$P(x) = 16x - 400$.

Find x so that $P(x) = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q1/5

$T(t) = 20 + 70(0.9)^t$.

Find $T(0)$.

$T(0) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q2/5

R(x) is the revenue in dollars from selling x items.

What does R(10) = 250 mean?

A. Selling 250 items brings in $10.

B. Each item costs $10.

C. The profit is $250.

D. Selling 10 items brings in $250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q3/5

True or false?

If C(x) = 400 + 9x, then C(0) = 400.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q4/5

True or false?

If $h(t) = -4.9t^2 + 30t + 2$, then $h(0) = 30$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSF-BF.A.1.b

FORWARD 1

Rewrite rational expressions such as average cost • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Write the average cost $A(x) = (400 + 9x) \div x$ as a constant plus a fraction.

$A(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q1/5

Assume $x \neq -1$. Write $(x^2 + 3x + 5) \div (x + 1)$ in the form $q(x) + r \div (x + 1)$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q2/5

Assume $x \neq 0$. Which expression equals $(6x^2 + 4x) \div (2x)$?

A. $3x + 4x$

B. $3x^2 + 2$

C. $6x + 2$

D. $3x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q3/5

True or false?

$(x^2 - 1) \div (x - 1) = x + 1$ for $x \neq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q4/5

True or false?

$(x + 8) \div x = 9$ for $x \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSF-BF.A.1.b

FORWARD 2

Graph and interpret an average-cost rational function • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

$A(x) = 9 + 400 \div x$ is the average cost per item, in dollars, for x items.

Give its horizontal asymptote and what it means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q1/5

$A(x) = 9 + 400 \div x$ is the average cost per item, in dollars, for x items.

Find A(100).

A(100) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q2/5

What is the vertical asymptote of $y = 400 \div x + 9$?

A. $x = 400$

B. $x = 9$

C. $y = 9$

D. $x = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q3/5

True or false?

As x increases, 9 + 400 ÷ x decreases toward 9.

A. True

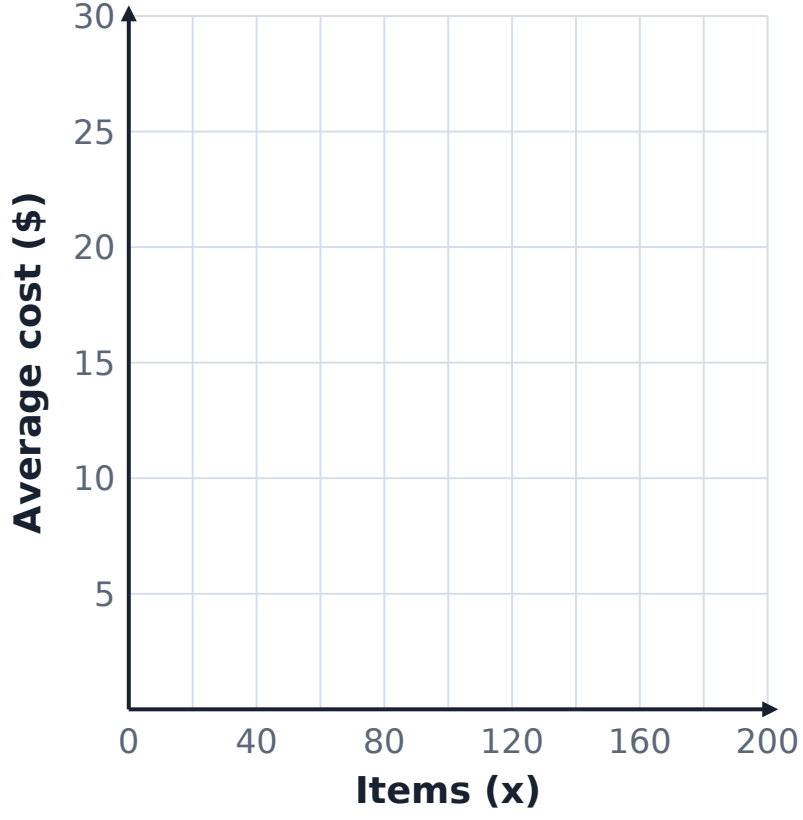
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q4/5

$A(x) = 9 + 400 \div x$ is the average cost per item, in dollars, for x items. Graph A for $20 \leq x \leq 200$, and draw its horizontal asymptote.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q5/5

Write arithmetic and geometric sequences recursively and with an explicit formula

MAIN	Algebra I main question	**HSF-BF.A.2**
BACKWARD 1	Evaluate recursively defined sequences	**HSF-IF.A.3**
BACKWARD 2	Generate a pattern from a rule and describe its features	**4.OA.C.5**
FORWARD 1	Sum geometric sequences	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to find a term number	**HSF-LE.A.4**

SET 50 • HSF-BF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

Write an explicit formula and a recursive formula for 7, 4, 1, −2, ...

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q1/5

Write an explicit formula and a recursive formula for 3, 12, 48, 192, ...

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q2/5

An arithmetic sequence has a(3) = 14 and a(8) = 39.

Find a(1) and the common difference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q3/5

A geometric sequence of real numbers has a(1) = 2 and a(4) = 54.

What is the common ratio?

A. 3

B. 18

C. 27

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q4/5

True or false?

The sequence 2, 4, 8, 14, ... is arithmetic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q5/5

SET 50 • HSF-BF.A.2

BACKWARD 1

Evaluate recursively defined sequences • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

$f(1) = 4$ and $f(n) = f(n - 1) + 6$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q1/5

$f(1) = 1$ and $f(n) = 2f(n - 1) + 1$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q2/5

$f(n) = 2n - 1$.

What is $f(10)$?

A. 20

B. 21

C. 19

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q3/5

True or false?

A recursive formula defines each term after the starting term(s) using one or more earlier terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q4/5

True or false?

$f(1) = 3$ and $f(n) = 2f(n - 1)$ give 3, 5, 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSF-BF.A.2

BACKWARD 2

Generate a pattern from a rule and describe its features • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

Rule: start at 5 and add 5.

Write the first 6 terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q1/5

Rule: start at 1 and add 4.

Explain why every term is odd.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q2/5

Rule: start at 2 and multiply by 2.

What is the 5th term?

A. 10

B. 16

C. 64

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q3/5

True or false?

With the rule "start at 3 and add 3," every term is a multiple of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q4/5

True or false?

With the rule "start at 10 and add 10," the 4th term is 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSF-BF.A.2

FORWARD 1

Sum geometric sequences • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 6 terms of 3, 12, 48, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q1/5

A job pays $1 on day 1, and the pay doubles each day.

What is the total pay for 10 days?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q2/5

What is 5 + 10 + 20 + 40 + 80?

A. 160

B. 80

C. 155

D. 150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q3/5

True or false?

The formula $a(1 - r^n) \div (1 - r)$ works for $r = 0.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q4/5

True or false?

$$2 + 6 + 18 + 54 = 2(3^4 - 1)$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSF-BF.A.2

FORWARD 2

Use logarithms to find a term number • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$a(n) = 3 \cdot 4^{n-1}$.

What is the smallest n for which a(n) > 10,000?

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

A sequence starts at 5 and grows by 20% per term: $a(n) = 5(1.2)^{n-1}$. Which term is the first to exceed 100? (log means base 10.)

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

Solve $5 \cdot 2^n = 600$. (log means base 10.)

A. n = 7

B. n = log 600 ÷ log 2 ≈ 9.23

C. n = log 120 ÷ log 2 ≈ 6.91

D. n = 120 ÷ 2 = 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

True or false?

If $4^k = 64$, then k = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

True or false?

If $3^n = 30$, then n = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

Model situations with arithmetic and geometric sequences and translate between recursive and explicit forms

MAIN	Algebra I main question	**HSF-BF.A.2**
BACKWARD 1	Evaluate sequences from recursive and explicit rules	**HSF-IF.A.3**
BACKWARD 2	Write a function or a recursive process from a situation	**HSF-BF.A.1.a**
FORWARD 1	Sum geometric sequences that model situations	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to find when a sequence reaches a value	**HSF-LE.A.4**

SET 81 • HSF-BF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

A gym charges $50 to join and $30 each month. a(n) is the total paid after n months, with a(1) = 80.

Write a recursive rule and an explicit rule for a(n).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q1/5

A ball bounces to 80% of its previous height. Its first bounce reaches 5 m.

Write an explicit rule and a recursive rule for the height h(n) of the nth bounce.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q2/5

Translate the explicit rule $a(n) = 7 + 4(n - 1)$, for $n \geq 1$, into a recursive rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q3/5

Which explicit rule matches $a(1) = 3$ and $a(n) = 2a(n - 1)$ for $n \geq 2$?

A. $a(n) = 3 \cdot 2^{n-1}$

B. $a(n) = 2 \cdot 3^{n-1}$

C. $a(n) = 3 \cdot 2^{n}$

D. $a(n) = 3 + 2(n - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q4/5

True or false?

The recursive rule $a(n) = a(n - 1) - 6$ needs a starting value, such as $a(1) = 50$, to define a sequence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 81 • MAIN • Q5/5

SET 81 • HSF-BF.A.2

BACKWARD 1

Evaluate sequences from recursive and explicit rules • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

$f(1) = 50$ and $f(n) = f(n - 1) - 6$ for $n \geq 2$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q1/5

$f(n) = 5(0.8)^{n-1}$.

Find f(3).

f(3) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q2/5

$f(1) = 2$ and $f(n) = 3f(n - 1) + 1$ for $n \geq 2$.

What is $f(3)$?

A. 7

B. 22

C. 19

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q3/5

True or false?

The domain of the sequence $a(n) = 2n + 1$, $n \geq 1$, is the positive integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q4/5

True or false?

$f(n) = n^2$ gives the sequence 1, 3, 5, 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • HSF-BF.A.2

BACKWARD 2

Write a function or a recursive process from a situation • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

A savings jar starts with $20 and gains $5 each week.

Write a function for the amount A after w weeks.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q1/5

A culture of 100 cells triples every day.

Write a function for the number of cells N after d days.

N = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q2/5

Which recursive process means "start at 10 and multiply by 1.1 each year"?

A. $a(0) = 10, a(n) = a(n - 1) + 1.1$

B. $a(n) = 10 + 1.1n$

C. $a(0) = 1.1, a(n) = 10a(n - 1)$

D. $a(0) = 10, a(n) = 1.1a(n - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q3/5

True or false?

Adding the same amount at each step gives a linear function of the step number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q4/5

True or false?

Multiplying by 2 at each step, starting at 1, gives f(n) = 2n.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 81 • BACKWARD 2 • Q5/5

SET 81 • HSF-BF.A.2

FORWARD 1

Sum geometric sequences that model situations • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

A ball's bounce heights are 5, 4, 3.2, ... meters (each 80% of the one before). Find the total of the first 5 bounce heights, to the nearest hundredth.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q1/5

You deposit $100 the first month and 3% more each month after that. Find the total deposited in 12 months, to the nearest cent.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q2/5

Which expression is the sum of the first 10 terms of $5(0.8)^{n-1}$?

A. $5(1 - 0.8^{10}) \div 0.2$

B. $5(0.8)^{10}$

C. $10(5)(0.8)$

D. $5 \div 0.2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q3/5

True or false?

The sum of the first n bounce heights 5, 4, 3.2, ... is always less than 25.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q4/5

True or false?

For a geometric series with ratio 2, adding one more term always doubles the sum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 81 • FORWARD 1 • Q5/5

SET 81 • HSF-BF.A.2

FORWARD 2

Use logarithms to find when a sequence reaches a value • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$h(n) = 5(0.8)^{n-1}$ is the height of the nth bounce, in meters. Which bounce is the first below 1 m? (log means base 10.)

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q1/5

$a(n) = 1{,}000(1.05)^n$ is a savings balance after n years. After how many whole years does it first exceed $2,000? (log means base 10.)

Years: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q2/5

Solve $3^n = 50$. (log means base 10.)

A. n = log 3 ÷ log 50

B. n = 50 ÷ 3

C. n = log 50 ÷ log 3 ≈ 3.56

D. n = 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S81-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q3/5

True or false?

$\log(0.8^n) = n \log 0.8$, and log 0.8 is negative. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q4/5

True or false?

If $2^n = 10$, then n is a whole number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 81 • FORWARD 2 • Q5/5

Identify the effect of replacing f(x) by f(x) + k, kf(x), f(kx), and f(x + k), and recognize even and odd functions

MAIN	Algebra I main question	**HSF-BF.B.3**
BACKWARD 1	Graph absolute value and square root functions	**HSF-IF.C.7.b**
BACKWARD 2	Read key features of quadratic graphs	**HSF-IF.C.7.a**
FORWARD 1	Describe a sequence of rigid transformations	**HSG-CO.A.5**
FORWARD 2	Transform sine and cosine graphs	**HSF-TF.B.5**

SET 51 • HSF-BF.B.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

$g(x) = (x - 4)^2 + 1$.

Describe how to get the graph of g from the graph of $y = x^2$.

Answer: _____

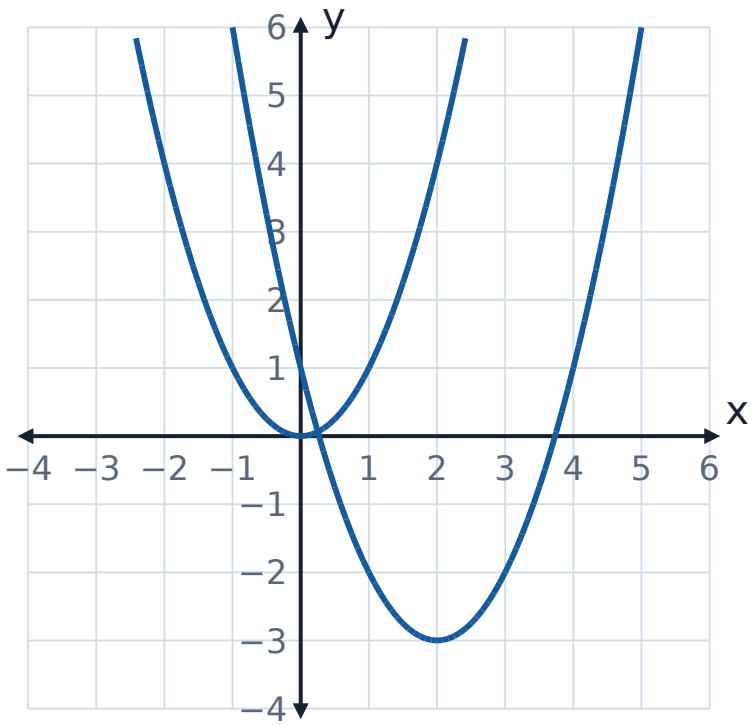
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q1/5

The graph shows $y = x^2$ and a translated graph $y = g(x)$.

Write $g(x)$.



$g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q2/5

$f(x) = x^2$. Compare $g(x) = 3f(x)$ and $h(x) = f(3x)$.

Which is a vertical stretch and which is a horizontal compression? Are g and h the same function?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q3/5

$f(x) = x^3$.

Is f even, odd, or neither?

A. Odd: $f(-x) = -f(x)$

B. Even: $f(-x) = f(x)$

C. Both even and odd

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q4/5

True or false?

The graph of $y = f(x + 2)$ is the graph of $y = f(x)$ shifted 2 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q5/5

SET 51 • HSF-BF.B.3

BACKWARD 1

Graph absolute value and square root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

What is the vertex of $y = |x + 3| - 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q1/5

Give two points on the graph of $y = \sqrt{(x - 1)}$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q2/5

What is the vertex of $y = -|x - 1| + 4$?

A. (1, 4)

B. (1, −4)

C. (4, 1)

D. (−1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q3/5

True or false?

$y = |x|$ has its minimum at (0, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q4/5

True or false?

The domain of $y = \sqrt{(x + 2)}$ is $x \geq 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSF-BF.B.3

BACKWARD 2

Read key features of quadratic graphs • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

Give the vertex of $y = (x - 2)^2 - 3$, and say whether it is a maximum or a minimum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q1/5

Find the x-intercepts of $y = x^2 - 9$.

x-intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q2/5

What is the y-intercept of $y = 2x^2 - 5x + 7$?

A. 7

B. 0

C. -5

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q3/5

True or false?

The graph of $y = -x^2 + 4$ opens downward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q4/5

True or false?

The vertex of $y = x^2 + 6$ is $(6, 0)$.

A. True

B. False

SET 51 • HSF-BF.B.3

FORWARD 1

Describe a sequence of rigid transformations • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5

Describe a transformation that maps the triangle with vertices (0, 0), (2, 0), (0, 1) to the triangle with vertices (3, 2), (5, 2), (3, 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q1/5

The point (2, 5) is reflected across the x-axis and then translated 4 units left.

Where does it end up?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q2/5

Which single transformation maps every (x, y) to (−x, −y)?

A. A translation

B. A rotation of 180° about the origin

C. A reflection across the y-axis

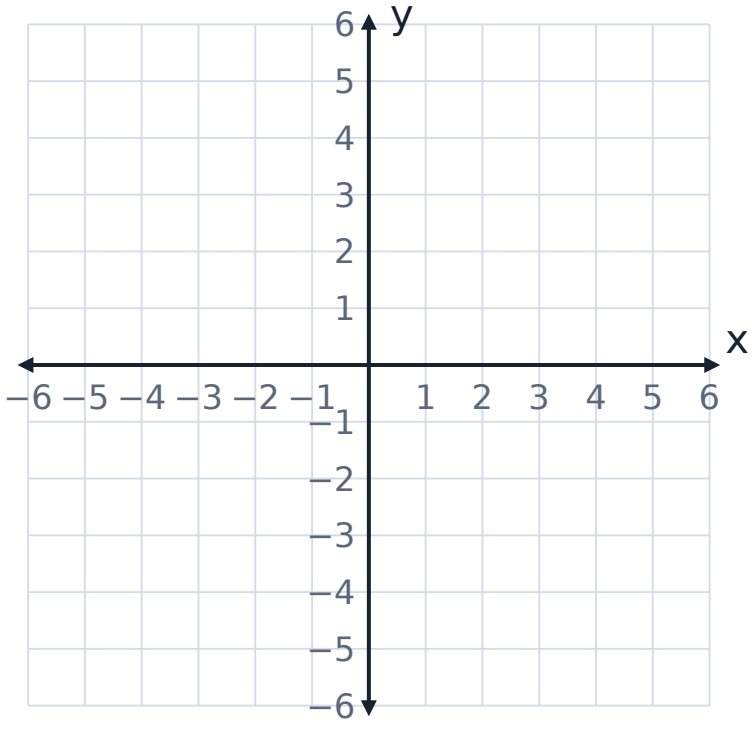
D. A reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q3/5

Triangle ABC has A(1, 1), B(4, 1), and C(1, 3). Reflect it across the y-axis, then translate the image 2 units down. Draw the final image and label its vertices.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q4/5

True or false?

Translating and then reflecting always gives the same result as reflecting and then translating.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSF-BF.B.3

FORWARD 2

Transform sine and cosine graphs • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

How is the graph of $y = \sin x + 3$ related to the graph of $y = \sin x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q1/5

Write a sine function with amplitude 2, midline $y = 1$, and period 2π whose graph passes through $(0, 1)$ and is rising there.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q2/5

How is $y = 4\sin x$ obtained from $y = \sin x$?

A. A shift 4 units right

B. A horizontal stretch by a factor of 4

C. A vertical stretch by a factor of 4

D. A shift 4 units up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q3/5

True or false?

$y = -\sin x$ is the reflection of $y = \sin x$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q4/5

True or false?

$y = \sin x - 2$ has midline $y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q5/5

Experiment with transformations of graphs and recognize even and odd functions from graphs and equations

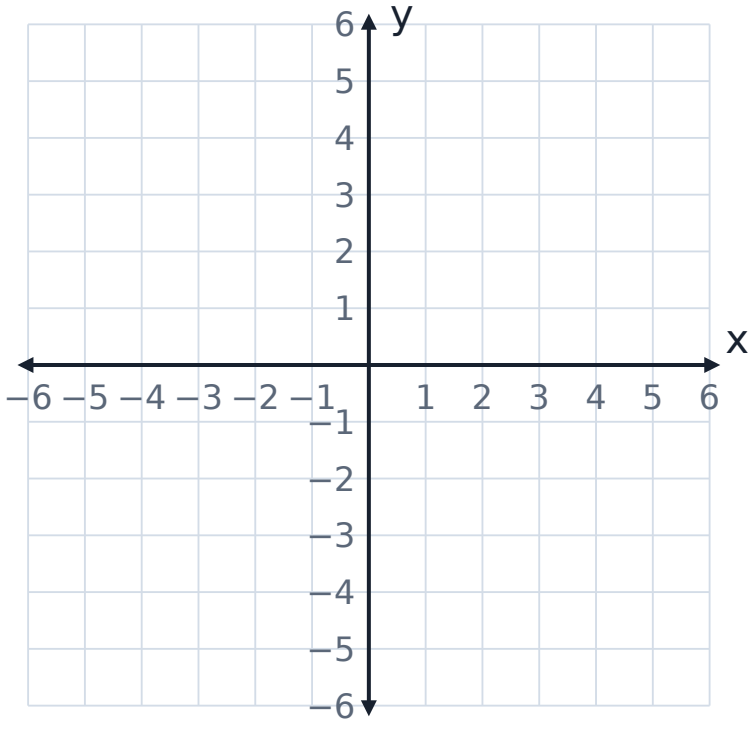
MAIN	Algebra I main question	**HSF-BF.B.3**
BACKWARD 1	Graph quadratic functions	**HSF-IF.C.7.a**
BACKWARD 2	Graph absolute value and root functions	**HSF-IF.C.7.b**
FORWARD 1	Draw rigid transformations of figures	**HSG-CO.A.5**
FORWARD 2	Transform sine and cosine graphs and recognize their symmetry	**HSF-TF.B.5**

SET 82 • HSF-BF.B.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

Graph $f(x) = x^2$ and $g(x) = (x - 3)^2 - 2$ on the same grid. Label both, and describe how g is related to f.

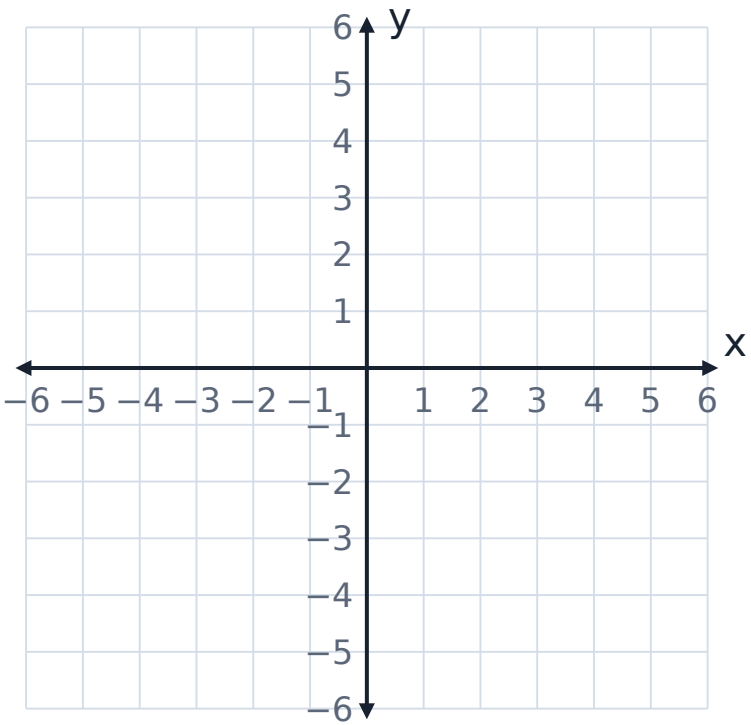


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q1/5

$f(x) = |x|$. Graph $g(x) = 2f(x)$ and $h(x) = f(2x)$ for $-3 \leq x \leq 3$, using a table or graphing technology. Are g and h the same function?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q2/5

Use a table or graphing technology to compare $f(x) = \sqrt{x}$ with $g(x) = \sqrt{(x - 4)} + 1$ at $x = 4$, 5, and 8. Describe the transformation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q3/5

The graph of f is shown.

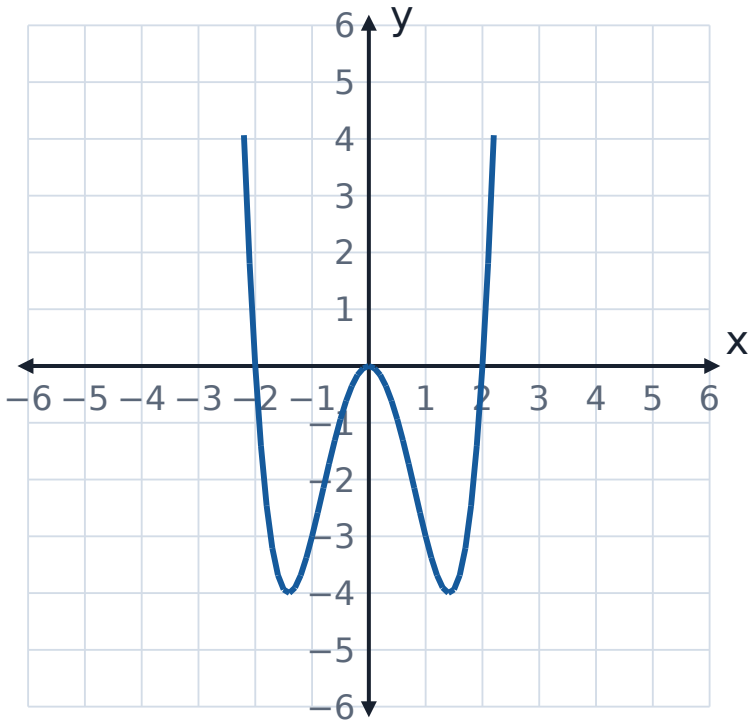
Is f even, odd, or neither?

A. Both even and odd

B. Even

C. Odd

D. Neither



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q4/5

True or false?

$f(x) = x^3 + x$ is an odd function, because $f(-x) = -f(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q5

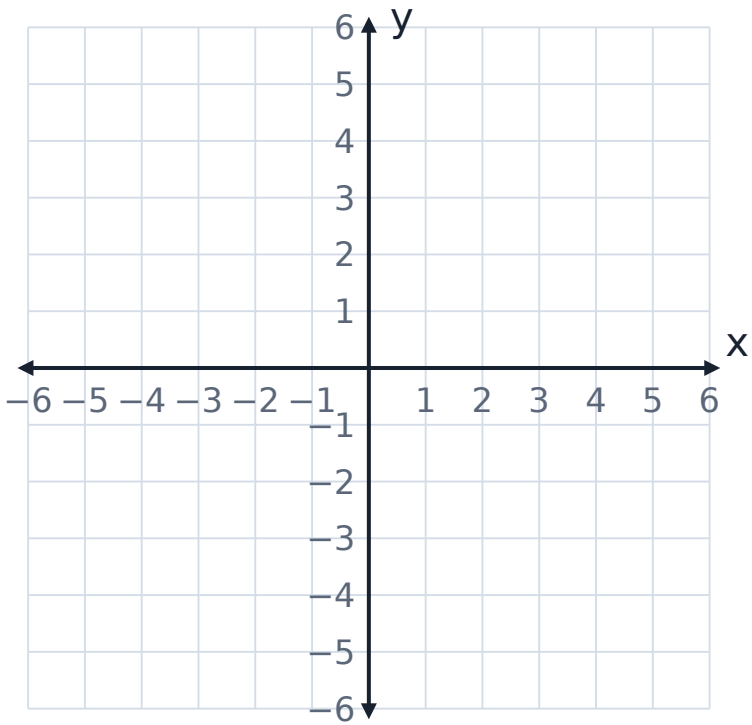
High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 82 • MAIN • Q5/5

SET 82 • HSF-BF.B.3

BACKWARD 1

Graph quadratic functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

Graph $y = x^2 - 4$, and mark its vertex and x-intercepts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q1/5

What is the vertex of $y = (x + 1)^2 + 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q2/5

What is the axis of symmetry of $y = x^2 - 6x + 1$?

A. $x = 6$

B. $x = 3$

C. $x = -3$

D. $x = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q3/5

True or false?

$y = -x^2 + 1$ has a maximum value of 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q4/5

True or false?

The x-intercepts of $y = x^2 + 1$ are 1 and -1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 82 • BACKWARD 1 • Q5/5

SET 82 • HSF-BF.B.3

BACKWARD 2

Graph absolute value and root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

What is the vertex of $y = |x - 4| + 1$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q1/5

Where does the graph of $y = \sqrt{(x + 2)} - 3$ start?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q2/5

Which function's graph is V-shaped?

A. $y = \sqrt[3]{(x + 1)}$

B. $y = |x + 1|$

C. $y = \sqrt{(x + 1)}$

D. $y = (x + 1)^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q3/5

True or false?

The graph of $y = \sqrt[3]{x}$ passes through $(-1, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q4/5

True or false?

The graph of $y = |x| - 2$ has vertex $(2, 0)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q5

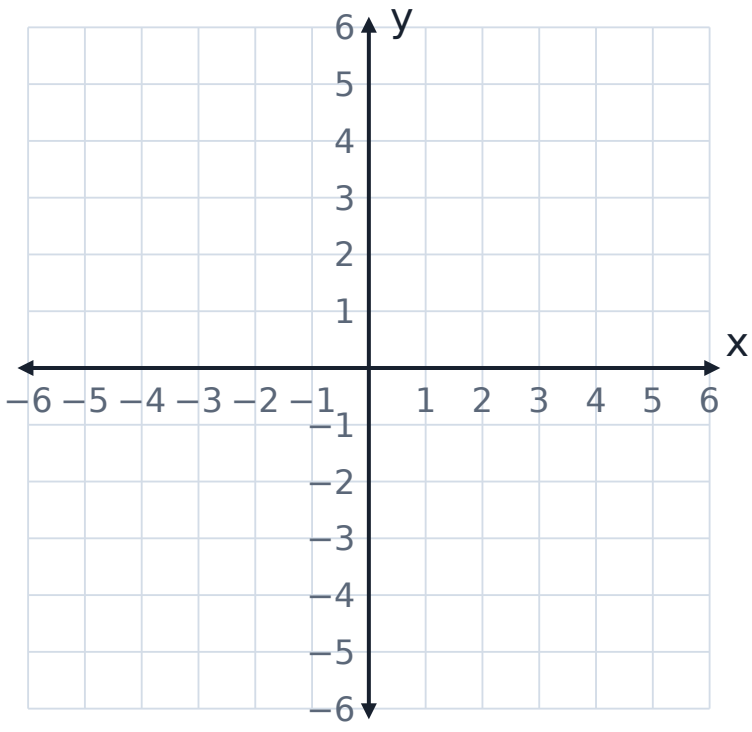
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 82 • BACKWARD 2 • Q5/5

SET 82 • HSF-BF.B.3

FORWARD 1

Draw rigid transformations of figures • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5

Translate triangle A(0, 0), B(2, 0), C(0, 3) by (x, y) → (x − 3, y + 1). Draw and label the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q1/5

The point (4, −1) is rotated 90° counterclockwise about the origin. Where does it go?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q2/5

Which sequence of transformations maps (x, y) to $(-x, y - 2)$?

A. Reflect across the x-axis, then translate 2 units down

B. Rotate 180°, then translate 2 units up

C. Reflect across the y-axis, then translate 2 units down

D. Translate 2 units left, then reflect across the y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q3/5

True or false?

A translation followed by another translation is a translation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q4/5

True or false?

Reflecting a figure across the same line twice moves it to a new position.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q5

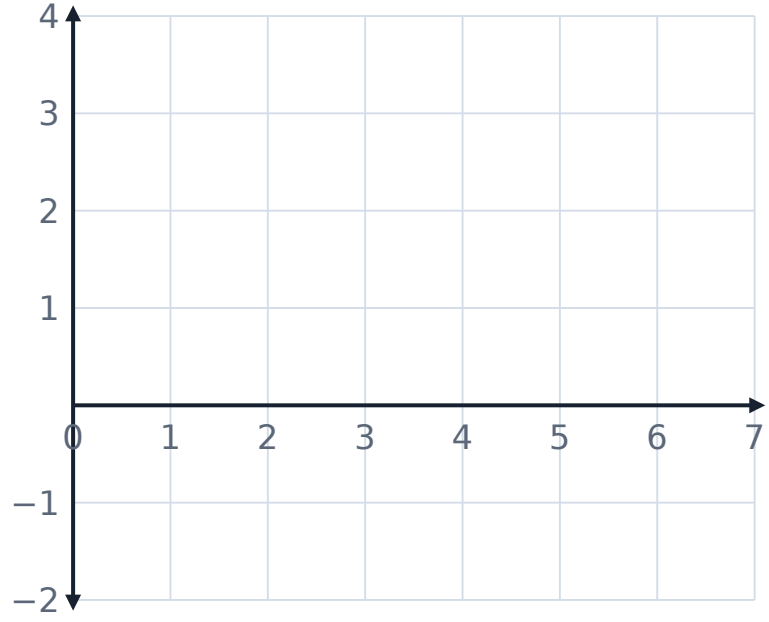
High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 82 • FORWARD 1 • Q5/5

SET 82 • HSF-BF.B.3

FORWARD 2

Transform sine and cosine graphs and recognize their symmetry • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

Sketch $y = 2\sin x + 1$ for $0 \le x \le 2\pi$ (about 6.28).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q1/5

Describe how the graph of $y = \sin(x - \pi \div 2)$ is related to the graph of $y = \sin x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q2/5

Which function has period π?

A. $y = \sin x + 2$

B. $y = \sin(x + 2)$

C. $y = 2\sin x$

D. $y = \sin(2x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q3/5

True or false?

$y = -\cos x$ is the reflection of $y = \cos x$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q4/5

True or false?

$y = \cos x$ is an odd function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 82 • FORWARD 2 • Q5/5

Solve f(x) = c for a simple function f and write an expression for the inverse

MAIN	Algebra I main question	**HSF-BF.B.4.a**
BACKWARD 1	Rearrange a formula for another variable	**HSA-CED.A.4**
BACKWARD 2	Use cube roots to solve $x^3 = p$	**8.EE.A.2**
FORWARD 1	Verify inverses by composition	**HSF-BF.B.4.b**
FORWARD 2	Restrict a domain to produce an inverse	**HSF-BF.B.4.d**

SET 52 • HSF-BF.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

$f(x) = 3x^3 - 1$. Solve $f(x) = 80$, and write an expression for the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q1/5

$f(x) = (2x + 3) \div (x - 4)$, for $x \neq 4$. Write an expression for the inverse.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q2/5

$F(C) = 1.8C + 32$ gives degrees Fahrenheit.

Write the inverse $C(F)$, and find C when $F = 77$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q3/5

$f(x) = 4x - 7$.

What is $f^{-1}(x)$?

A. $(x - 7) \div 4$

B. $(x + 7) \div 4$

C. $x \div 4 + 7$

D. $4x + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q4/5

True or false?

If $f(x) = x \div 3 + 2$, then $f^{-1}(x) = 3x - 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q5/5

SET 52 • HSF-BF.B.4.a

BACKWARD 1

Rearrange a formula for another variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve $y = 4x - 7$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q1/5

Solve $P = 2l + 2w$ for l.

$l =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q2/5

Solve $d = rt$ for t. ($r \neq 0$)

A. $t = r \div d$

B. $t = d - r$

C. $t = d \div r$

D. $t = dr$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q3/5

True or false?

Solving $y = x^3$ for x gives $x = \sqrt[3]{y}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q4/5

True or false?

Solving $y = 5x$ for x gives $x = 5y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSF-BF.B.4.a

BACKWARD 2

Use cube roots to solve $x^3 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve $x^3 = 27$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q1/5

Evaluate $\sqrt[3]{125}$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q2/5

What is $\sqrt[3]{64}$?

A. 21.3

B. 4

C. 8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q3/5

True or false?

$\sqrt[3]{(-27)} = -3$

A. True

B. False

True or false?

$\sqrt{16} = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSF-BF.B.4.a

FORWARD 1

Verify inverses by composition • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

Show that $f(x) = 2x^3$ and $g(x) = \sqrt[3]{(x \div 2)}$ are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q1/5

$f(x) = (x + 1) \div (x - 1)$.

Find $f(f(3))$.

$f(f(3)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q2/5

Which function is its own inverse?

A. $f(x) = 1 \div x$

B. $f(x) = 2x$

C. $f(x) = x + 1$

D. $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F1-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q3/5

True or false?

If f(g(x)) = x on the domain of g and g(f(x)) = x on the domain of f, then g is the inverse of f.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q4/5

True or false?

$f(x) = x + 3$ and $g(x) = x - 3$ are not inverses, because $f(g(1)) = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSF-BF.B.4.a

FORWARD 2

Restrict a domain to produce an inverse • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d

$f(x) = x^2 + 2$. Restrict f to the largest interval that contains $x = 1$ on which f is one-to-one, and give the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q1/5

Explain why f(x) = |x| has no inverse on all real numbers.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q2/5

$f(x) = (x + 4)^2$.

What is the largest interval containing $x = 0$ on which f is one-to-one?

A. All real numbers

B. $-8 \leq x \leq 0$

C. $x \geq 0$

D. $x \geq -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^2$ restricted to $x \leq 0$ has the inverse $f^{-1}(x) = -\sqrt{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q4/5

True or false?

$f(x) = x^2 + 1$ is one-to-one on all real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 52 • FORWARD 2 • Q5/5

Prove that linear functions grow by equal differences and exponential functions grow by equal factors over equal intervals

MAIN	Algebra I main question	**HSF-LE.A.1.a**
BACKWARD 1	Tell linear from nonlinear functions	**8.F.A.3**
BACKWARD 2	Compare average rates of change over intervals	**HSF-IF.B.6**
FORWARD 1	Derive the formula for the sum of a finite geometric series	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to turn equal factors into equal differences	**HSF-BF.B.5**

SET 53 • HSF-LE.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

$f(x) = 3x + 5$. Show that f grows by equal differences over equal intervals: find $f(x + h) - f(x)$ in simplest form, and explain why it does not depend on x.

$f(x + h) - f(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q1/5

$g(x) = 4 \cdot 2^x$. Show that g grows by equal factors over equal intervals: find $g(x + h) \div g(x)$ in simplest form, and explain why it does not depend on x.

$g(x + h) \div g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q2/5

A table shows x = 0, 2, 4, 6 and y = 5, 11, 17, 23.

Is the relationship linear or exponential? Explain using equal intervals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q3/5

A table shows x = 0, 1, 2, 3 and y = 80, 40, 20, 10.

Which description is correct?

A. Linear: y decreases by 40 each time

B. Exponential: y is multiplied by $\frac{1}{2}$ each time x increases by 1

C. Linear: y decreases by 20 each time

D. Neither linear nor exponential

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q4/5

True or false?

A table with y-values 2, 5, 10, 17 for x = 0, 1, 2, 3 is linear.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 53 • MAIN • Q5/5

SET 53 • HSF-LE.A.1.a

BACKWARD 1

Tell linear from nonlinear functions • Grade 8 • Functions • 8.F.A.3

Is $y = x^2 + 1$ linear? Use the points (0, 1), (1, 2), and (2, 5) to explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q1/5

Is $y = 4 - 2x$ linear? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q2/5

Which function is linear?

A. $y = 2^x$

B. $y = x^2$

C. $y = 0.5x - 3$

D. $y = 1 \div x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q3/5

True or false?

The graph of a linear function is a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q4/5

True or false?

The area of a square, $A = s^2$, is a linear function of s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSF-LE.A.1.a

BACKWARD 2

Compare average rates of change over intervals • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = 3x + 5$.

Find the average rate of change from $x = 0$ to $x = 10$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q1/5

$g(x) = 4 \cdot 2^x$.

Find the average rate of change from $x = 0$ to $x = 1$, and from $x = 1$ to $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q2/5

What is the average rate of change of $5 \cdot 3^x$ from $x = 0$ to $x = 2$?

A. 15

B. 20

C. 40

D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q3/5

True or false?

An exponential growth function has a larger average rate of change on later intervals of the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 53 • BACKWARD 2 • Q4/5

True or false?

The average rate of change of 2^x from $x = 0$ to $x = 3$ is 2.

A. True

B. False

SET 53 • HSF-LE.A.1.a

FORWARD 1

Derive the formula for the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Let $S = 1 + 2 + 4 + \ldots + 2^9$. Show that $2S - S = 2^{10} - 1$, and find S.

$S =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q1/5

Use $S = a(1 - r^n) \div (1 - r)$ to find the sum of the first 6 terms of 1, 3, 9, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q2/5

When deriving the sum formula, why do we multiply S by r?

A. To double the sum

B. So that $S - rS$ cancels all but two terms

C. To make every term equal

D. To make r equal 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q3/5

True or false?

For $r \neq 1$, $a + ar + \ldots + ar^{n-1} = a(1 - r^n) \div (1 - r)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 53 • FORWARD 1 • Q4/5

True or false?

$1 + 3 + 9 + 27 = 3^4$

A. True

B. False

SET 53 • HSF-LE.A.1.a

FORWARD 2

Use logarithms to turn equal factors into equal differences • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

$g(x) = 4 \cdot 2^x$.

Show that $\log_2 g(x) = x + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q1/5

$y = 10^x$.

What is log y? (log means base 10.)

log y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q2/5

The values y = 1, 10, 100, 1,000 occur at x = 0, 1, 2, 3.

What are the values of log y? (log means base 10.)

A. 0, 1, 2, 3

B. 0, 10, 20, 30

C. 1, 2, 3, 4

D. 1, 10, 100, 1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q3/5

True or false?

For $a > 0$ and $b > 0$, taking the logarithm of $a \cdot b^x$ gives a linear function of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q4/5

True or false?

$\log(100 \cdot 10^x) = 100x$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 53 • FORWARD 2 • Q5/5

Recognize situations in which a quantity changes at a constant rate per unit interval

MAIN	Algebra I main question	**HSF-LE.A.1.b**
BACKWARD 1	Find the constant of proportionality	**7.RP.A.2.b**
BACKWARD 2	Write a linear function from a rate and an initial value	**8.F.B.4**
FORWARD 1	Relate arc length and sector area to the central angle — nearest r…	**HSG-C.B.5**
FORWARD 2	Use radian measure as arc length on a circle — nearest related	**HSF-TF.A.1**

SET 54 • HSF-LE.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b

A pool fills at a constant rate. At 2 p.m. it holds 1,500 gallons; at 5 p.m. it holds 3,300 gallons.

What is the rate, and how much water was in it at noon?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q1/5

A plant is 6 cm tall on day 0 and grows 0.8 cm each day.

How tall is it on day 15?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q2/5

A taxi charges the same amount for every mile.

Explain why the fare for the miles driven is a linear function of the miles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q3/5

Which situation is modeled by a linear function?

A. A cell population doubles each hour.

B. A tank loses 4 liters each minute.

C. A loan grows 4% each month.

D. A car loses 15% of its value each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q4/5

True or false?

A savings account that earns the same dollar amount of interest each year grows linearly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 54 • MAIN • Q5/5

SET 54 • HSF-LE.A.1.b

BACKWARD 1

Find the constant of proportionality • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

$y = 2.5x$ gives the cost of x pounds of apples.

What is the constant of proportionality?

Constant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q1/5

A proportional table has x = 2, 4, 6 and y = 7, 14, 21.

What is the unit rate?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q2/5

A cyclist rides 12 miles in 3 hours at a constant rate.

What is the constant of proportionality in miles per hour?

A. 15

B. 36

C. $\frac{1}{4}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q3/5

True or false?

In $y = 9x$, y increases by 9 whenever x increases by 1.

A. True

B. False

True or false?

A proportional relationship through (5, 20) has constant of proportionality 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSF-LE.A.1.b

BACKWARD 2

Write a linear function from a rate and an initial value • Grade 8 • Functions • 8.F.B.4

A line passes through (2, 1,500) and (5, 3,300).

Find its slope and its y-intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q1/5

A phone battery is at 80% and drops 5 percentage points per hour.

Write a function B(t) for the charge after t hours.

B(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q2/5

Which function has rate of change 0.8 and initial value 6?

A. $y = 6x + 0.8$

B. $y = 0.8x + 6$

C. $y = 6.8x$

D. $y = 6(0.8)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q3/5

True or false?

A function with initial value 10 and rate of change -2 is $y = 10 - 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q4/5

True or false?

The line through (0, 4) and (2, 10) has rate of change 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSF-LE.A.1.b

FORWARD 1

Relate arc length and sector area to the central angle • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Geometry standard. Connection: Arc length grows by a constant amount (the radius) for each radian, a constant-rate situation.

In a circle of radius 5 cm, by how much does the arc length increase for each 1-radian increase in the central angle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

A circle has radius 10.

Find the arc lengths for central angles of 0.5, 1, and 1.5 radians.

Arc lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

The area of a sector is $A = \frac{1}{2}r^2\theta$, and $r = 4$.

By how much does A increase when θ increases by 1 radian?

A. 4

B. 2

C. 16

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

For a fixed radius, arc length is a linear function of the central angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Doubling the radius while keeping the angle the same doubles the area of a sector.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 54 • HSF-LE.A.1.b

FORWARD 2

Use radian measure as arc length on a circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Algebra II standard. Connection: Radian measure is arc length per unit of radius, continuing the constant-rate arc length of FORWARD 1.

A wheel of radius 0.3 m turns through 4π radians.

How far does a point on its rim travel? Give the exact distance and the distance to the nearest hundredth of a meter.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q1/5 • NEAREST RELATED

A central angle cuts off an arc of length 6 on a circle of radius 2.

What is its radian measure?

Radians: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q2/5 • NEAREST RELATED

A wheel turns at 3 radians per second.

Through about how many degrees does it turn in 2 seconds?

A. 6°

B. 172°

C. 1,080°

D. 344°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

One radian is the angle that cuts off an arc as long as the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A rotation of $-\pi \div 2$ radians is a quarter turn counterclockwise.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 54 • FORWARD 2 • Q5/5 • NEAREST RELATED

Recognize situations in which a quantity grows or decays by a constant percent rate per unit interval

MAIN	Algebra I main question	**HSF-LE.A.1.c**
BACKWARD 1	Read growth and decay rates from exponential functions	**HSF-IF.C.8.b**
BACKWARD 2	Solve percent problems	**7.RP.A.3**
FORWARD 1	Find times in percent-rate models with logarithms	**HSF-LE.A.4**
FORWARD 2	Use natural logarithms with continuous growth	**HSF-BF.B.5**

SET 55 • HSF-LE.A.1.c

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

A car worth $24,000 loses 15% of its value each year.

Write a function V(t), and find the value after 3 years to the nearest dollar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q1/5

Explain why a quantity that grows by 6% each year grows by equal factors over equal intervals.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q2/5

A town's population increases by 2.5% per year.

By what factor does it grow in 2 years? What is the percent increase over 2 years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q3/5

Which situation is exponential?

A. A worker earns $15 each hour.

B. A medicine dose decreases by 20% each hour.

C. A tank gains 3 liters each second.

D. A runner covers 200 meters each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-M-Q4**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q4/5

True or false?

A 10% increase followed by a 10% decrease returns a quantity to its original value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 55 • MAIN • Q5/5

SET 55 • HSF-LE.A.1.c

BACKWARD 1

Read growth and decay rates from exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

$V = 24{,}000(0.85)^t$.

What is the yearly percent change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q1/5

$P = 500(1.025)^t$.

Is this growth or decay, and at what rate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q2/5

Which function shows 4% decay per year?

A. $y = 300(1.04)^t$

B. $y = 300(0.04)^t$

C. $y = 300 - 0.04t$

D. $y = 300(0.96)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q3/5

True or false?

$y = 50(1.5)^t$ grows by 50% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q4/5

True or false?

$y = 80(0.75)^t$ decreases by 75% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSF-LE.A.1.c

BACKWARD 2

Solve percent problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A $24,000 car loses 15% of its value.

What is its new value?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q1/5

What is 8% sales tax on $45?

Tax: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q2/5

What is the price of a $60 item after a 25% discount?

A. $15

B. $45

C. $75

D. $35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q3/5

True or false?

20% of 150 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q4/5

True or false?

A 50% increase of 40 is 60, and a 50% decrease of 60 is 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSF-LE.A.1.c

FORWARD 1

Find times in percent-rate models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

$V(t) = 24{,}000(0.85)^t$.

When is V = 12,000? Round to the nearest tenth of a year.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q1/5

Solve $1.025^t = 2$. Round to the nearest tenth.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q2/5

Which expression gives the half-life of $y = (0.9)^t$? (log means base 10.)

A. 0.9 ÷ 2

B. 0.5 ÷ 0.9

C. log 0.9 ÷ log 0.5

D. log 0.5 ÷ log 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q3/5

True or false?

$\log(0.85^t) = t \log 0.85$ (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 55 • FORWARD 1 • Q4/5

True or false?

If $2^t = 5$, then $t = 2.5$.

A. True

B. False

SET 55 • HSF-LE.A.1.c

FORWARD 2

Use natural logarithms with continuous growth • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

Write the inverse of $y = e^x$.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q1/5

Solve $e^{0.05t} = 3$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q2/5

What is $\ln(e^4)$? (ln means base e.)

A. 4e

B. 4

C. 1

D. e

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q3/5

True or false?

e raised to the power ln 7 equals 7. (ln means base e.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 55 • FORWARD 2 • Q4/5

True or false?

ln 1 = 1 (ln means base e.)

A. True

B. False

Construct linear and exponential functions from a graph, a description, two points, or a table

MAIN	Algebra I main question	**HSF-LE.A.2**
BACKWARD 1	Write a linear function from two points or a table	**8.F.B.4**
BACKWARD 2	Write formulas for arithmetic and geometric sequences	**HSF-BF.A.2**
FORWARD 1	Build a sinusoidal model from a description	**HSF-TF.B.5**
FORWARD 2	Find a quadratic through three points with a matrix equation	**HSA-REI.C.8**

SET 56 • HSF-LE.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the linear function whose graph passes through (2, 11) and (6, 23).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q1/5

Write the exponential function $y = ab^x$ $(b > 0)$ whose graph passes through $(1, 6)$ and $(3, 54)$.

$y =$ _____

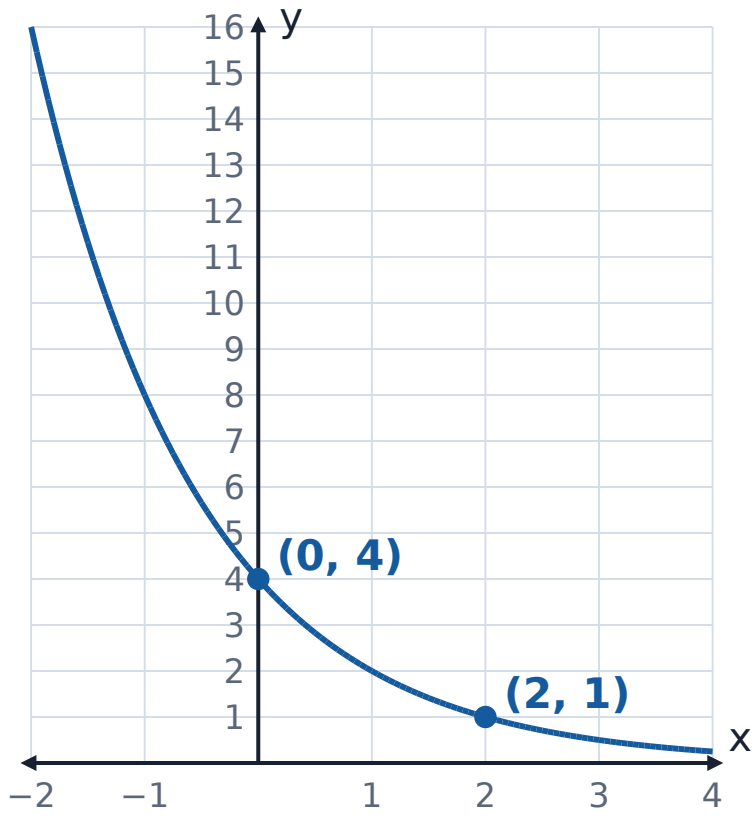
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q2/5

The graph of an exponential function $y = ab^x$ passes through the marked points.

Write the function.



$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q3/5

A table shows x = 0, 1, 2, 3 and y = 1.5, 3, 6, 12.

Which function fits?

A. $y = 2(1.5)^x$

B. $y = 1.5x + 1.5$

C. $y = 3x$

D. $y = 1.5(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q4/5

True or false?

The linear function with $f(0) = 10$ that decreases by 2 for each unit increase in x is $f(x) = 10(0.8)^x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 56 • MAIN • Q5/5

SET 56 • HSF-LE.A.2

BACKWARD 1

Write a linear function from two points or a table • Grade 8 • Functions • 8.F.B.4

Write the linear function through (0, −3) and (4, 5).

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q1/5

A table shows x = 1, 2, 3 and y = 10, 7, 4.

Write the linear function.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q2/5

Which line passes through $(-1, 2)$ and $(1, 6)$?

A. $y = -2x + 4$

B. $y = 2x + 4$

C. $y = 2x + 2$

D. $y = 4x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q3/5

True or false?

The line through (0, 0) and (3, 6) is $y = 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q4/5

True or false?

The line through (1, 1) and (2, 3) is $y = 2x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSF-LE.A.2

BACKWARD 2

Write formulas for arithmetic and geometric sequences • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

A geometric sequence has a(1) = 2 and ratio 3.

Write an explicit formula.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q1/5

Write an explicit formula for 10, 7, 4, ...

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q2/5

Which is a recursive formula for 6, 3, 1.5, ...?

A. $a(n) = 6 - 3n$

B. $a(1) = 6, a(n) = 0.5a(n - 1)$

C. $a(1) = 6, a(n) = 2a(n - 1)$

D. $a(1) = 6, a(n) = a(n - 1) - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q3/5

True or false?

An arithmetic sequence is a linear function on the positive integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q4/5

True or false?

The geometric sequence 5, 10, 20, ... has ratio 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSF-LE.A.2

FORWARD 1

Build a sinusoidal model from a description • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

A tide has high water 10 ft and low water 2 ft, with period 12 hours and high tide at t = 0.

Write a cosine model h(t).

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q1/5

A sinusoid has maximum 10 and minimum 2.

What are its amplitude and midline?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q2/5

A model $y = \cos(Bt)$ has period 12.

What is B?

A. 12

B. $\pi \div 12$

C. $\pi \div 6$

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q3/5

True or false?

A sinusoidal model with maximum 8 and minimum 0 has amplitude 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q4/5

True or false?

$y = 3\cos(2t)$ has period 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSF-LE.A.2

FORWARD 2

Find a quadratic through three points with a matrix equation • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8

$y = ax^2 + bx + c$ passes through (0, 1), (1, 2), and (2, 7).

Write the system for a, b, and c as a matrix equation $AX = B$, with X the column [a, b, c]. Give A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q1/5

A has rows [0 0 1], [1 1 1], and [4 2 1], X is the column [a, b, c], and B is the column [1, 2, 7]. Solve AX = B to find the parabola $y = ax^2 + bx + c$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q2/5

A parabola $y = ax^2 + bx + c$ passes through (0, 1), (1, 2), and (2, 7). It is written as $AX = B$ with X the column $[a, b, c]$.

Which row of A comes from the point (2, 7)?

A. [4 2 1]

B. [2 4 1]

C. [4 2 7]

D. [2 2 1]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q3/5

True or false?

Three points with different x-values that are not on one line determine exactly one parabola $y = ax^2 + bx + c$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q4/5

True or false?

A parabola $y = ax^2 + bx + c$ through (0, 1), (1, 2), and (2, 7) gives a matrix equation $AX = B$ in which A has 2 rows.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 56 • FORWARD 2 • Q5/5

Observe that exponential growth eventually exceeds linear, quadratic, and polynomial growth

MAIN	Algebra I main question	**HSF-LE.A.3**
BACKWARD 1	Tell equal differences from equal factors	**HSF-LE.A.1.a**
BACKWARD 2	Compare two functions given in different forms	**8.F.A.2**
FORWARD 1	Compare the growth of polynomial functions	**HSF-IF.C.7.c**
FORWARD 2	Compare logarithmic growth with other growth	**HSF-BF.B.5**

SET 57 • HSF-LE.A.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3

$f(x) = 100x$ and $g(x) = 2^x$.

What is the smallest whole number $x \geq 1$ for which $g(x) > f(x)$?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q1/5

$f(x) = x^2$ and $g(x) = 1.5^x$.

Which is larger at $x = 10$? At $x = 20$? What happens for larger x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q2/5

Plan A pays $1,000 per day. Plan B pays 1 cent on day 1, and its daily pay doubles each day.

On which day does Plan B's daily pay first exceed Plan A's?

Day: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q3/5

Which function eventually has the greatest values as x increases?

A. $y = 50x^2$

B. $y = 1.01^x$

C. $y = x^3 + 500$

D. $y = 1{,}000x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.3 • Set 57 • MAIN • Q4/5

True or false?

Because $3x^2 > 2^x$ at $x = 5$, $3x^2$ stays greater than 2^x for all $x > 5$.

A. True

B. False

SET 57 • HSF-LE.A.3

BACKWARD 1

Tell equal differences from equal factors • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

A table shows x = 0, 1, 2, 3 and y = 3, 6, 12, 24.

Is it linear or exponential? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q1/5

A table shows x = 0, 1, 2, 3 and y = 3, 6, 9, 12.

Is it linear or exponential? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q2/5

Which function grows by equal factors over equal intervals?

A. $y = 7 + 1.1x$

B. $y = 7(1.1)^x$

C. $y = 7x$

D. $y = 1.1x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q3/5

True or false?

Linear functions grow by equal differences over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q4/5

True or false?

$y = 5x^2$ grows by equal factors over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSF-LE.A.3

BACKWARD 2

Compare two functions given in different forms • Grade 8 • Functions • 8.F.A.2

Function A is $y = 4x + 1$. Function B has $x = 0, 1, 2$ and $y = 3, 5, 7$.

Which has the greater rate of change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q1/5

Function A is $y = -x + 6$. Function B is the line through $(0, 2)$ and $(2, 8)$.

Which has the greater y-intercept?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q2/5

$f(x) = 3x - 2$. Function g has $g(0) = 1$ and $g(1) = 5$, and it is linear.

Which statement is true?

A. f has the greater rate of change.

B. f has the greater initial value.

C. g has the greater rate of change and the greater initial value.

D. f and g have the same rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q3/5

True or false?

A linear function with slope 5 increases faster than one with slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q4/5

True or false?

$y = 2x + 10$ has greater values than $y = 3x$ for every x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSF-LE.A.3

FORWARD 1

Compare the growth of polynomial functions • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

For which $x > 0$ is $x^3 > 100x^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q1/5

Which grows faster for large x, x^4 or $1{,}000x^3$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q2/5

As $x \to \infty$, what does $f(x) = x^4 - 50x^3$ do?

A. $f(x) \to -\infty$

B. $f(x) \to -50$

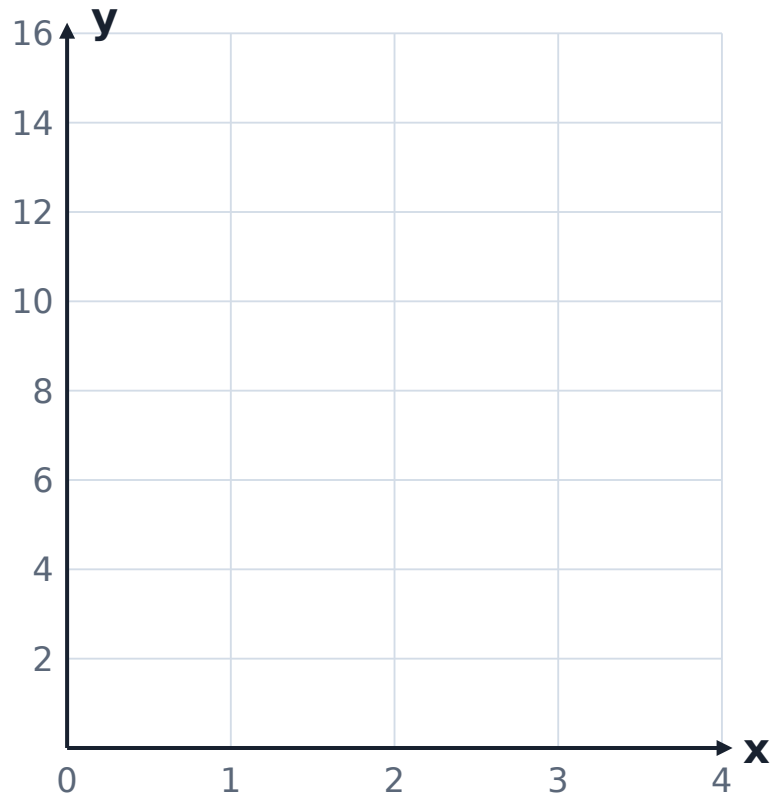
C. $f(x) \to \infty$

D. $f(x) \to 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q3/5

Graph $y = x^2$ and $y = 0.5x^3$ for $0 \le x \le 4$ on the same grid. Mark where they meet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q4/5

True or false?

For every large x, $0.001x^3$ is less than $1{,}000x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSF-LE.A.3

FORWARD 2

Compare logarithmic growth with other growth • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

Find $\log_2 1{,}024$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q1/5

By what factor must x be multiplied for log x to increase by 1? (log means base 10.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q2/5

Which is greatest when x = 1,000,000? (log means base 10; ln means base e.)

A. $\sqrt{x}$

B. $\log_2 x$

C. $\ln x$

D. $\log x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q3/5

True or false?

Eventually, log x grows more slowly than any linear function with positive slope. (log means base 10.)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 57 • FORWARD 2 • Q4/5

True or false?

$\log_2 x = 100$ when $x = 200$.

A. True

B. False

Interpret the parameters of a linear or exponential function in context

MAIN	Algebra I main question	**HSF-LE.B.5**
BACKWARD 1	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**
BACKWARD 2	Interpret the rate and initial value of a linear function	**8.F.B.4**
FORWARD 1	Interpret the parameters of a sinusoidal model	**HSF-TF.B.5**
FORWARD 2	Interpret the mean and standard deviation of a normal model — n…	**HSS-ID.A.4**

SET 58 • HSF-LE.B.5

MAIN

Algebra I main question • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

$C(m) = 0.35m + 40$ is the cost in dollars of renting a car for one day and driving m miles.

What do 0.35 and 40 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q1/5

$P(t) = 1{,}200(1.04)^t$ is a town's population t years after 2020.

What do 1,200 and 1.04 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q2/5

$A(t) = 80(0.5)^{t/6}$ is the milligrams of a medicine left in the body after t hours.

What do 80, 0.5, and 6 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q3/5

$V(t) = 18{,}000 - 1{,}500t$ is a machine's value in dollars after t years.

What does $-1{,}500$ mean?

A. The machine loses $1,500 of value each year.

B. The machine loses 1,500% of its value each year.

C. The machine was bought for $1,500.

D. The machine is worth $18,000 after 1,500 years.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q4/5

True or false?

In $y = 250(0.92)^t$, the value decreases by 92% each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 58 • MAIN • Q5/5

SET 58 • HSF-LE.B.5

BACKWARD 1

Interpret the parts of an expression in context • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A plumber charges 75 + 60h dollars for h hours of work.

What does 75 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q1/5

In $3{,}000(1.05)^t$, what does the factor 1.05 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q2/5

A box with a square base of side s and height h has surface area $2s^2 + 4sh$.

What does 4sh represent?

A. The volume

B. The area of the four side faces

C. The perimeter of the base

D. The area of the top and bottom

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q3/5

True or false?

In 500 − 20w, the term −20w shows a decrease of 20 for each unit of w.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q4/5

True or false?

In 1.5x + 7, the coefficient of x is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSF-LE.B.5

BACKWARD 2

Interpret the rate and initial value of a linear function • Grade 8 • Functions • 8.F.B.4

Cost y = 12x + 30 for x hours of bike rental.

What is the initial value, and what does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q1/5

The water in a tank is $y = 500 - 25t$ gallons after t minutes.

What does -25 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q2/5

A plant's height is a linear function of time. It is 8 cm on day 0 and 20 cm on day 4.

What is the rate of change?

A. 5 cm per day

B. 20 cm per day

C. 8 cm per day

D. 3 cm per day

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q3/5

True or false?

In $y = 40x + 100$, 100 is the value of y when $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 58 • BACKWARD 2 • Q4/5

True or false?

In d = 65t, 65 is the starting distance.

A. True

B. False

SET 58 • HSF-LE.B.5

FORWARD 1

Interpret the parameters of a sinusoidal model • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

$T(t) = 15 + 10\sin(\pi t \div 12)$ is the temperature in °C t hours after 8 a.m.

What do 15 and 10 mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q1/5

$T(t) = 15 + 10\sin(\pi t \div 12)$ is the temperature in °C t hours after 8 a.m.

What is the period of T, and what does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q2/5

$h(t) = 1.5\cos(\pi t \div 6) + 3$ is the depth of water in meters.

What is the greatest depth?

A. 4.5 m

B. 3 m

C. 6 m

D. 1.5 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q3/5

True or false?

In $y = A \sin(Bx) + D$, D gives the midline.

A. True

B. False

True or false?

The amplitude of $y = 1.5\cos(\pi t \div 6) + 3$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSF-LE.B.5

FORWARD 2

Interpret the mean and standard deviation of a normal model • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Interpreting the parameters of a function in context leads to interpreting the parameters of a normal model.

Battery life is normal with mean 400 hours and standard deviation 20 hours.

What does the 20 tell you?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q1/5 • NEAREST RELATED

Battery life is normal with mean 400 hours and standard deviation 20 hours.

About what percent of batteries last between 360 and 440 hours?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 58 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which change makes a normal curve wider?

A. Increasing the standard deviation

B. Increasing the mean

C. Decreasing the standard deviation

D. Decreasing the mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Changing the mean of a normal model shifts the curve left or right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 58 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

In a normal model with mean 50 and standard deviation 5, the value 60 is 10 standard deviations above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 58 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 59 • HSS-ID.A.1 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Represent data with dot plots, histograms, and box plots

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Display numerical data in plots	**6.SP.B.4**
BACKWARD 2	Compute measures of center and variation	**6.SP.B.5.c**
FORWARD 1	Decide whether a normal model fits a distribution	**HSS-ID.A.4**
FORWARD 2	Decide whether results are consistent with a model by simulation…	**HSS-IC.A.2**

SET 59 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a dot plot of the numbers of books read by 10 students:

2, 3, 3, 4, 4, 4, 5, 5, 6, 9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q1/5

Find the five-number summary of 2, 3, 3, 4, 4, 4, 5, 5, 6, 9. Use the median of each half (lower five and upper five values) as Q1 and Q3.

Summary: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q2/5

For the data 2, 3, 3, 4, 4, 4, 5, 5, 6, 9 (Q1 and Q3 are the medians of the lower five and upper five values), use the 1.5 × IQR rule to decide whether 9 is an outlier.

Answer: _____

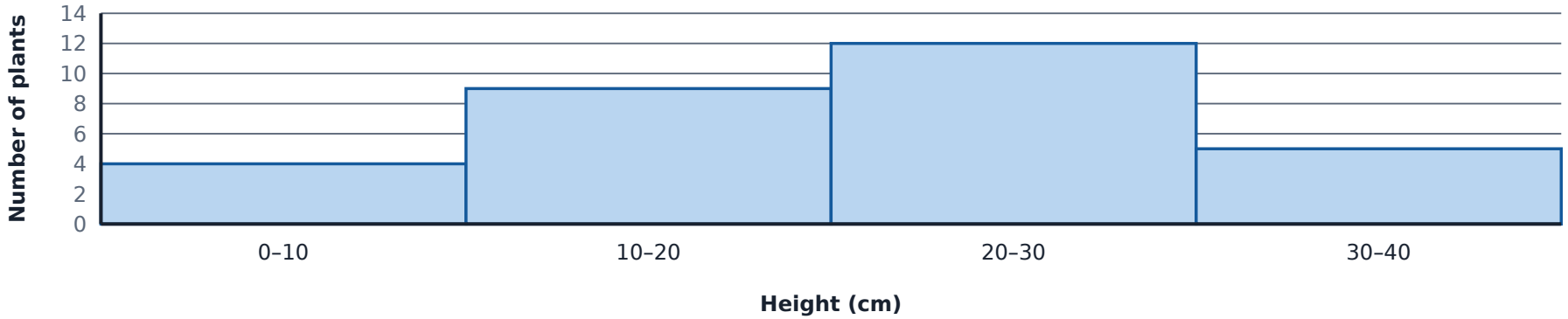
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q3/5

The histogram shows the heights of 30 plants. Each bar includes its lower edge but not its upper edge.

How many plants are at least 20 cm tall?



A. 21 **B.** 17 **C.** 5 **D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-M-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q4/5

True or false?

A box plot shows the mean of a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 59 • MAIN • Q5/5

SET 59 • HSS-ID.A.1

BACKWARD 1

Display numerical data in plots • Grade 6 • Statistics & Probability • 6.SP.B.4

Which display shows every data value, a dot plot or a box plot?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q1/5

Make a dot plot of 1, 2, 2, 3, 3, 3, 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q2/5

Which display groups data values into intervals?

A. Box plot

B. Dot plot

C. Histogram

D. Ordered list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q3/5

True or false?

A box plot is made from the five-number summary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q4/5

True or false?

A histogram shows each individual data value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSS-ID.A.1

BACKWARD 2

Compute measures of center and variation • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 3, 8, 5, 10, 4.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q1/5

Find the mean of 3, 8, 5, 10, 4.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q2/5

Q1 = 12 and Q3 = 20.

What is the interquartile range?

A. 32

B. 8

C. 4

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q3/5

True or false?

The range of 2, 7, 9, 15 is 13.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q4/5

True or false?

The median of 1, 2, 3, 100 is 26.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSS-ID.A.1

FORWARD 1

Decide whether a normal model fits a distribution • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

A data set is strongly skewed to the right.

Is a normal model appropriate? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q1/5

Scores are normal with mean 75 and standard deviation 8.

What score is 1 standard deviation above the mean?

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q2/5

Values are normal with mean 20 and standard deviation 4.

About what percent are between 12 and 28?

A. 99.7%

B. 50%

C. 68%

D. 95%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 59 • FORWARD 1 • Q3/5

True or false?

In a normal distribution, the mean and the median are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 59 • FORWARD 1 • Q4/5

True or false?

About 50% of a normal distribution lies within 1 standard deviation of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSS-ID.A.1

FORWARD 2

Decide whether results are consistent with a model by simulation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Simulation results are summarized with the data displays of the MAIN set and then compared with a model.

In 100 simulated sets of 10 fair coin flips, 8 or more heads happened in 5 sets.

Estimate the probability of 8 or more heads in 10 fair flips.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 59 • FORWARD 2 • Q1/5 • NEAREST RELATED

A spinner model says red comes up 25% of the time. In 200 spins, red came up 52 times.

What proportion is that? Is it consistent with the model?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 59 • FORWARD 2 • Q2/5 • NEAREST RELATED

A model says a coin lands heads 50% of the time.

Which result most strongly questions the model?

A. 52 heads in 100 flips

B. 6 heads in 10 flips

C. 95 heads in 100 flips

D. 48 heads in 100 flips

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 59 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A result that rarely happens in simulations of a model is evidence against the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 59 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

If a coin lands heads 6 times in 10 flips, the coin must be unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 59 • FORWARD 2 • Q5/5 • NEAREST RELATED

Make histograms from data

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Display data in a histogram	**6.SP.B.4**
BACKWARD 2	Report the number of observations	**6.SP.B.5.a**
FORWARD 1	Use a histogram's shape to decide whether a normal model fits	**HSS-ID.A.4**
FORWARD 2	Use a histogram of simulated results to judge a model	**HSS-IC.A.2**

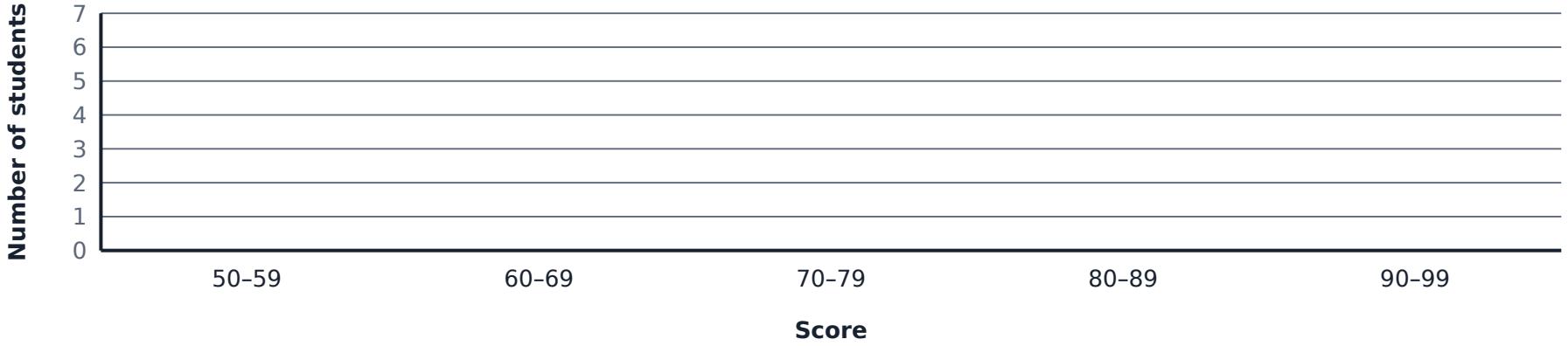
SET 76 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a histogram of these 20 test scores. Use the intervals 50–59, 60–69, 70–79, 80–89, and 90–99.

52, 58, 61, 64, 67, 69, 70, 72, 74, 75, 77, 78, 81, 83, 85, 86, 88, 91, 94, 97



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q1/5

Using the intervals 50–59, 60–69, 70–79, 80–89, and 90–99 for the scores 52, 58, 61, 64, 67, 69, 70, 72, 74, 75, 77, 78, 81, 83, 85, 86, 88, 91, 94, 97:

how many scores are in the 70–79 interval, and which interval has the tallest bar?

Answer: _____

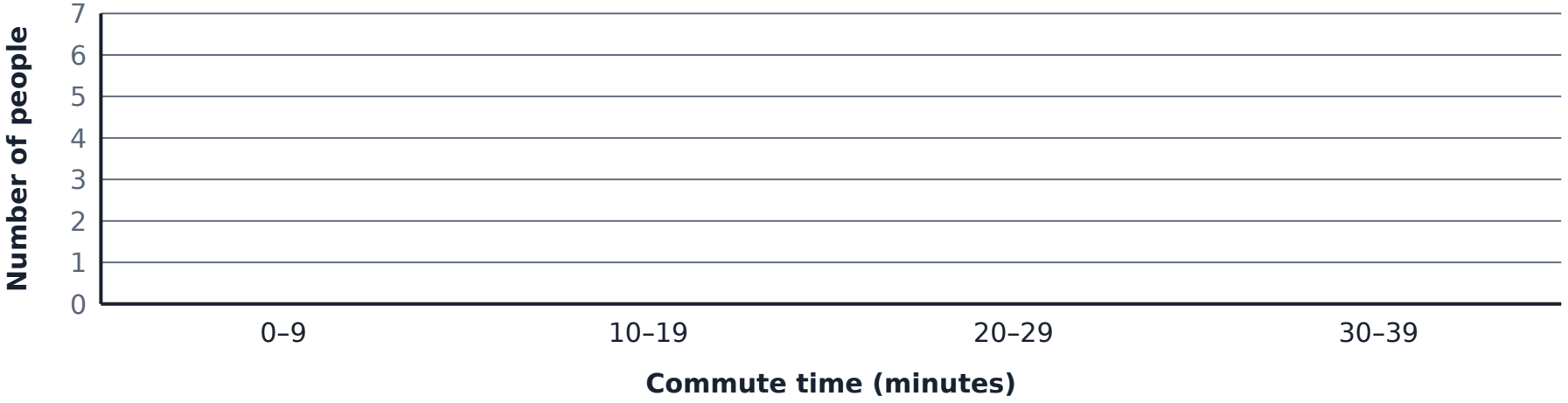
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q2/5

Make a histogram of these 16 commute times, in minutes. Use the intervals 0–9, 10–19, 20–29, and 30–39.

4, 8, 12, 14, 15, 18, 19, 21, 22, 22, 25, 28, 29, 33, 36, 38



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q3/5

A histogram uses the intervals 10–19 and 20–29 for whole-number data.

In which bar is the value 20 counted?

A. 20–29

B. 10–19

C. Both

D. Neither

True or false?

Using wider intervals for the same data gives a histogram with more bars.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q5

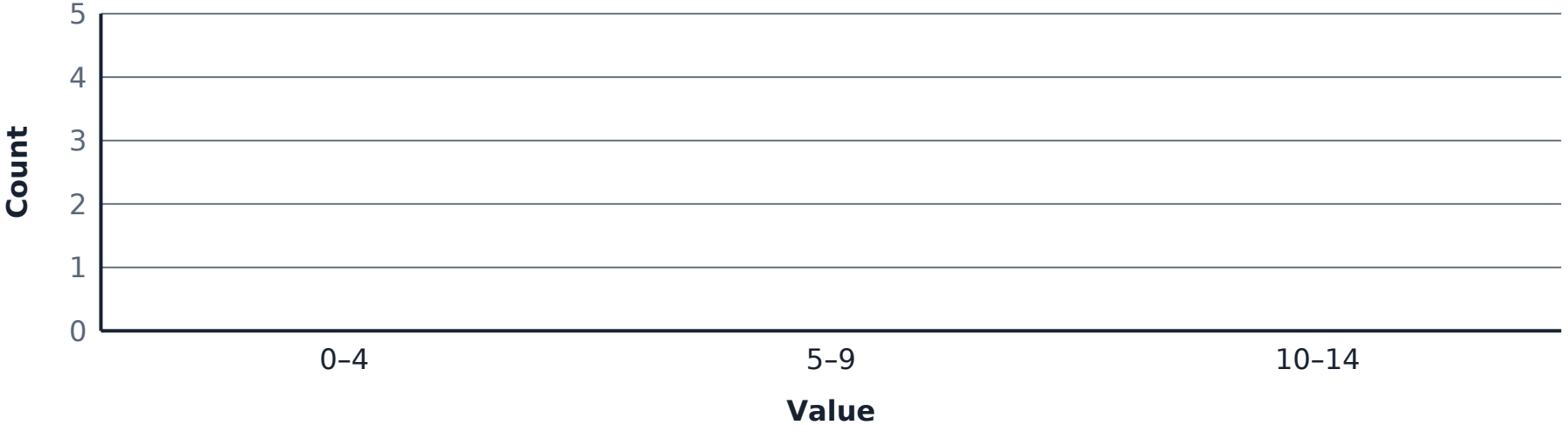
High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 76 • MAIN • Q5/5

SET 76 • HSS-ID.A.1

BACKWARD 1

Display data in a histogram • Grade 6 • Statistics & Probability • 6.SP.B.4

Make a histogram of 1, 3, 4, 6, 7, 8, 9, 12 with the intervals 0–4, 5–9, and 10–14.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q1/5

A histogram has bars of heights 3, 5, and 2. How many data values does it show?

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q2/5

Which data are best shown in a histogram?

A. The heights of 200 students

B. Each student's favorite color

C. One test score

D. The names of 5 cities

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q3/5

True or false?

In a histogram, the height of a bar shows how many values are in that interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q4/5

True or false?

A histogram is the best display for categories such as favorite colors.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 76 • BACKWARD 1 • Q5/5

SET 76 • HSS-ID.A.1

BACKWARD 2

Report the number of observations • Grade 6 • Statistics & Probability • 6.SP.B.5.a

How many observations are in the data set 4, 7, 7, 9, 10, 12?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q1/5

A histogram has bars of heights 2, 4, 6, 5, and 3. How many observations does it show?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q2/5

A survey recorded the number of pets owned by each of 25 students. How many observations are there?

A. 1

B. 25

C. 100

D. The total number of pets

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q3/5

True or false?

The number of observations is the number of data values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q4/5

True or false?

If 8 students each report 2 siblings, there are 16 observations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 76 • BACKWARD 2 • Q5/5

SET 76 • HSS-ID.A.1

FORWARD 1

Use a histogram's shape to decide whether a normal model fits • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

A histogram of 1,000 adult heights is single-peaked and symmetric, with mean 170 cm and standard deviation 8 cm. About how many heights are between 162 cm and 178 cm?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 76 • FORWARD 1 • Q1/5

A histogram of household incomes is strongly skewed right. Would a normal model give good estimates of percents? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q2/5

Which histogram shape suggests that a normal model may fit?

A. Two separate peaks

B. Flat (uniform)

C. Single-peaked and symmetric

D. Strongly skewed to the left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q3/5

True or false?

In a normal model, about 95% of values are within 2 standard deviations of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 76 • FORWARD 1 • Q4/5

True or false?

Any histogram with one peak is normal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 76 • FORWARD 1 • Q5/5

SET 76 • HSS-ID.A.1

FORWARD 2

Use a histogram of simulated results to judge a model • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

100 simulated trials of 20 fair coin flips give a histogram of the number of heads centered near 10. What does the histogram show?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 76 • FORWARD 2 • Q1/5

In that simulation, 2 of the 100 trials had 15 or more heads. Estimate the probability of 15 or more heads in 20 fair flips.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 76 • FORWARD 2 • Q2/5

A spinner is claimed to land on blue 50% of the time. In 50 spins it landed on blue 12 times. In 200 simulated sets of 50 spins with a 50% chance, 12 or fewer blue never happened.

What do you conclude?

A. Nothing can be concluded.

B. The result supports the 50% claim.

C. The spinner lands on blue exactly 24% of the time.

D. The result is not consistent with the 50% claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S76-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 76 • FORWARD 2 • Q3/5

True or false?

A simulation repeats a random process many times to show which results are typical under a model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 76 • FORWARD 2 • Q4/5

True or false?

If a result never happens in 200 simulated trials, it is impossible under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.2 • Set 76 • FORWARD 2 • Q5/5

Make box plots from data

MAIN	Algebra I main question	**HSS-ID.A.1**
BACKWARD 1	Draw and read box plots	**6.SP.B.4**
BACKWARD 2	Find medians, quartiles, and ranges	**6.SP.B.5.c**
FORWARD 1	Estimate the quartiles of a normal distribution	**HSS-ID.A.4**
FORWARD 2	Compare two treatments with box plots and a randomization test	**HSS-IC.B.5**

SET 77 • HSS-ID.A.1

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Make a box plot of 3, 5, 6, 7, 8, 9, 10, 12, 15. (Use the median of each half, not counting the middle value, as Q1 and Q3.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q1/5

Find the five-number summary of 12, 15, 15, 18, 20, 22, 25, 30. (Q1 and Q3 are the medians of the lower four and upper four values.)

Summary: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q2/5

Make a box plot of 12, 15, 15, 18, 20, 22, 25, 30. (Q1 and Q3 are the medians of the lower four and upper four values.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q3/5

In a box plot, what does the length of the box show?

A. The interquartile range

B. The mean

C. The range

D. The median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q4/5

True or false?

In a box plot whose whiskers reach the minimum and maximum, each whisker covers about a quarter of the data values.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 77 • MAIN • Q5/5

SET 77 • HSS-ID.A.1

BACKWARD 1

Draw and read box plots • Grade 6 • Statistics & Probability • 6.SP.B.4

Draw a box plot with minimum 2, Q1 4, median 5, Q3 8, and maximum 10.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q1/5

A box plot has Q1 = 20 and Q3 = 35. What is the interquartile range?

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q2/5

Which part of a box plot marks the median?

A. The line inside the box

B. The left end of the box

C. The right whisker

D. The left whisker

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q3/5

True or false?

A box plot shows the minimum and the maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q4/5

True or false?

A box plot shows how many data values there are.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 77 • BACKWARD 1 • Q5/5

SET 77 • HSS-ID.A.1

BACKWARD 2

Find medians, quartiles, and ranges • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 4, 9, 2, 7, 5, 8.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q1/5

The ordered data are 2, 4, 5, 7, 8, 9. Find Q1, the median of the lower half 2, 4, 5.

Q1 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q2/5

What is the range of 14, 3, 9, 21?

A. 21

B. 18

C. 7

D. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q3/5

True or false?

The interquartile range is Q3 − Q1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q4/5

True or false?

The median of 1, 3, 100 is 34.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • BACKWARD 2 • Q5/5

SET 77 • HSS-ID.A.1

FORWARD 1

Estimate the quartiles of a normal distribution • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

For a normal distribution, the quartiles are about $\mu - 0.67\sigma$ and $\mu + 0.67\sigma$. Estimate Q1 and Q3 for mean 100 and standard deviation 15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 77 • FORWARD 1 • Q1/5

Using those quartiles, estimate the interquartile range of that normal distribution.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 77 • FORWARD 1 • Q2/5

In a normal distribution, about what percent of values are between Q1 and Q3?

A. 25%

B. 95%

C. 50%

D. 68%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 77 • FORWARD 1 • Q3/5

True or false?

A box plot of data from a normal distribution is roughly symmetric about the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 77 • FORWARD 1 • Q4/5

True or false?

For normal data, the median is far from the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 77 • FORWARD 1 • Q5/5

SET 77 • HSS-ID.A.1

FORWARD 2

Compare two treatments with box plots and a randomization test • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Students were randomly assigned to a new study method (group A) or the usual method (group B). Box plots show a median of 78 for A and 70 for B, with similar spreads. In 1,000 random regroupings, a difference in means at least as large as observed happened 30 times.

Using a 5% threshold, is the difference significant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q1/5

Why is it useful to draw side-by-side box plots of the two groups before running the test?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q2/5

In another randomized comparison, random regroupings matched or beat the observed difference 22% of the time.

Using a 5% threshold, what is the conclusion?

A. The difference is significant.

B. The methods are proved to have the same effect.

C. The new method is proved to work.

D. The difference is not significant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S77-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q3/5

True or false?

In a randomized comparison, a difference that rarely appears in random regroupings is evidence of a treatment effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q4/5

True or false?

Overlapping box plots prove that two treatments have the same effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 77 • FORWARD 2 • Q5/5

SET 60 • HSS-ID.A.2 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Compare the center and spread of two or more data sets

MAIN	Algebra I main question	**HSS-ID.A.2**
BACKWARD 1	Tell measures of center from measures of variation	**6.SP.A.3**
BACKWARD 2	Compare two distributions using center and variability	**7.SP.B.3**
FORWARD 1	Use simulation to decide whether a difference between treatment…	**HSS-IC.B.5**
FORWARD 2	Evaluate reports based on data	**HSS-IC.B.6**

SET 60 • HSS-ID.A.2

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Class A scores: 70, 75, 80, 85, 90. Class B scores: 60, 70, 80, 90, 100.

Compare the centers and the spreads.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q1/5

Find the population standard deviation (divide by n) of 2, 4, 4, 4, 5, 5, 7, 9.

Standard deviation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q2/5

Store X's daily sales have median 50 and IQR 10. Store Y's have median 45 and IQR 25.

Which store typically sells more? Which is more consistent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q3/5

Which measure of spread is best for skewed data with outliers?

A. Range

B. Standard deviation

C. Mean

D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q4/5

True or false?

Two data sets with the same mean must have the same standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 60 • MAIN • Q5/5

SET 60 • HSS-ID.A.2

BACKWARD 1

Tell measures of center from measures of variation • Grade 6 • Statistics & Probability • 6.SP.A.3

Which describes a typical value of a data set, the median or the IQR?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q1/5

What is the range of the data set 4, 4, 4, 4?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q2/5

Which is a measure of variation?

A. Median

B. Mode

C. Mean

D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q3/5

True or false?

A measure of center summarizes all of the values with a single number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q4/5

True or false?

The range is a measure of center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSS-ID.A.2

BACKWARD 2

Compare two distributions using center and variability • Grade 7 • Statistics & Probability • 7.SP.B.3

Two classes have mean scores 78 and 84, and each has MAD 3.

How many MADs apart are the means?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q1/5

Team A's mean height is 170 cm and Team B's is 180 cm; each has MAD 5 cm.

Express the difference of the means as a multiple of the MAD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q2/5

Two means differ by 12, and each data set has MAD 4.

How many MADs is the difference?

A. 48

B. 12

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q3/5

True or false?

When two distributions have little variability, a small difference in their centers is easier to notice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q4/5

True or false?

If two data sets overlap, their means must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSS-ID.A.2

FORWARD 1

Use simulation to decide whether a difference between treatments is significant • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Plants were randomly assigned to a fertilizer group or a control group with no fertilizer. The fertilizer group averaged 24 cm and the control group 20 cm. To test chance alone, the same heights were randomly regrouped 1,000 times, and a difference of 4 cm or more (first group minus second) happened 12 times.

Using a 5% threshold, is the difference significant? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q1**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 60 • FORWARD 1 • Q1/5

Why are subjects randomly assigned to treatments in an experiment?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 60 • FORWARD 1 • Q2/5

In a regrouping simulation, a difference as large as the observed one happened 400 times out of 1,000.

What does that suggest?

A. The treatment definitely works.

B. The difference could easily be due to chance.

C. The treatment definitely does not work.

D. The experiment had no control group.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q3/5

True or false?

In a randomized experiment, a small simulated probability for the observed difference is evidence that the treatment had an effect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.5 • Set 60 • FORWARD 1 • Q4/5

True or false?

In an experiment, the control group receives the new treatment being tested.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.5 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSS-ID.A.2

FORWARD 2

Evaluate reports based on data • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

An ad says "9 out of 10 dentists recommend Brand X," based on 10 dentists chosen by Brand X.

What is the flaw?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q1/5

An online poll of a website's visitors finds 80% support a policy.

Can the result be applied to all adults? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 60 • FORWARD 2 • Q2/5

Which sample is most likely to represent a school's students?

A. The first 50 students to arrive

B. The 50 students in the chess club

C. A random sample of 50 students from the school roster

D. 50 volunteers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions

HSS-IC.B.6 • Set 60 • FORWARD 2 • Q3/5

True or false?

A larger random sample generally gives a smaller margin of error.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.B.6 • Set 60 • FORWARD 2 • Q4/5

True or false?

A survey with many volunteers is free of bias.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions
HSS-IC.B.6 • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSS-ID.A.3 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Interpret differences in shape, center, and spread, accounting for outliers

MAIN	Algebra I main question	**HSS-ID.A.3**
BACKWARD 1	Describe a distribution by its center, spread, and shape	**6.SP.A.2**
BACKWARD 2	Choose measures of center and spread that fit the data	**6.SP.B.5.d**
FORWARD 1	Use z-scores to judge how unusual a value is	**HSS-ID.A.4**
FORWARD 2	Distinguish a population parameter from a sample statistic — nea…	**HSS-IC.A.1**

SET 61 • HSS-ID.A.3

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Salaries at a small company, in thousands of dollars: 40, 42, 45, 45, 48, 50, 250.

Find the mean and the median. Which better describes a typical salary?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q1/5

Salaries at a small company, in thousands of dollars: 40, 42, 45, 45, 48, 50, 250.

If the salary of 250 is removed, what happens to the mean and to the median?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q2/5

A histogram of wait times has a long tail on the right.

Describe its shape and the usual relationship between its mean and median.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q3/5

Which data set has the largest standard deviation?

A. 28, 29, 30, 31, 32

B. 25, 30, 30, 30, 35

C. 10, 20, 30, 40, 50

D. 30, 30, 30, 30, 30

True or false?

An outlier affects the median more than the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 61 • MAIN • Q5/5

SET 61 • HSS-ID.A.3

BACKWARD 1

Describe a distribution by its center, spread, and shape • Grade 6 • Statistics & Probability • 6.SP.A.2

Data: 1, 2, 2, 3, 3, 3, 4, 4, 5.

Describe the shape of the distribution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q1/5

Which value in 2, 3, 3, 3, 4, 15 is unusual?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q2/5

A dot plot has most values on the left and a few values far to the right.

What is its shape?

A. Skewed right

B. Symmetric

C. Uniform

D. Skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q3/5

True or false?

A distribution can be described by its center, its spread, and its shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q4/5

True or false?

A symmetric distribution must have an outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSS-ID.A.3

BACKWARD 2

Choose measures of center and spread that fit the data • Grade 6 • Statistics & Probability • 6.SP.B.5.d

For data with an extreme outlier, should you report the mean or the median as the center? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q1/5

For symmetric data with no outliers, which measures of center and spread are usually reported?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q2/5

Home prices in a town include one mansion.

Which measure best describes a typical price?

A. The mean

B. The median

C. The maximum

D. The range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q3/5

True or false?

The median is not changed much by a single extreme value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q4/5

True or false?

The IQR changes greatly when the largest value is made even larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSS-ID.A.3

FORWARD 1

Use z-scores to judge how unusual a value is • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

A distribution has mean 50 and standard deviation 4.

What is the z-score of 58?

$z =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

A score of 92 on a test with mean 80 and standard deviation 6, or a score of 85 on a test with mean 70 and standard deviation 10: which is better relative to its test?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

In a normal distribution, which z-score is the most unusual?

A. -3.1

B. -0.5

C. 1.2

D. 2.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

True or false?

A z-score of 0 means the value equals the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

True or false?

A z-score of -2 is above the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

FORWARD 2

Distinguish a population parameter from a sample statistic • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

⚑ No standard in a later course directly continues this skill at this step. This branch uses the nearest related Probability & Statistics standard. Connection: Centers and spreads computed from data are sample statistics that estimate population parameters.

What is the difference between a population and a sample?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 61 • FORWARD 2 • Q1/5 • NEAREST RELATED

A random sample of 200 students has a mean of 7.1 hours of sleep.

What does 7.1 estimate?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 61 • FORWARD 2 • Q2/5 • NEAREST RELATED

A parameter describes

A. a graph

B. one individual

C. a population

D. a sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 61 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

Different random samples from the same population can give different sample means.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 61 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A sample statistic always equals the population parameter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Making Inferences & Justifying Conclusions HSS-IC.A.1 • Set 61 • FORWARD 2 • Q5/5 • NEAREST RELATED

SET 62 • HSS-ID.B.5 • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Summarize categorical data in two-way frequency tables and interpret relative frequencies

MAIN	Algebra I main question	**HSS-ID.B.5**
BACKWARD 1	Read two-way tables for association	**8.SP.A.4**
BACKWARD 2	Write a part of a whole as a percent	**7.RP.A.3**
FORWARD 1	Use a two-way table to find conditional probabilities and test inde…	**HSS-CP.A.4**
FORWARD 2	Explain conditional probability and independence in everyday lang..	**HSS-CP.A.5**

SET 62 • HSS-ID.B.5

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

The table shows a survey of 200 students.

What percent of all the students play a sport?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q1/5

What percent of Grade 10 students play a sport? What percent of Grade 9 students do?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q2/5

Is there an association between grade and playing a sport? Explain.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q3/5

What is the joint relative frequency of "Grade 9 and does not play a sport"?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. 50%

B. 57.9%

C. 55%

D. 27.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q4/5

True or false?

Of the students who play a sport, more than half are in Grade 10.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 62 • MAIN • Q5/5

SET 62 • HSS-ID.B.5

BACKWARD 1

Read two-way tables for association • Grade 8 • Statistics & Probability • 8.SP.A.4

12 of 20 boys and 9 of 15 girls in a class like math.

How many students in all like math?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q1/5

12 of 20 boys and 9 of 15 girls in a class like math.

Is there an association between gender and liking math? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q2/5

What are the row and column totals of a two-way table called?

A. Joint frequencies

B. Conditional frequencies

C. Marginal frequencies

D. Outliers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q3/5

True or false?

A two-way table shows data on two categorical variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q4/5

True or false?

In a two-way table, every row must have the same total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSS-ID.B.5

BACKWARD 2

Write a part of a whole as a percent • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Write 45 out of 100 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q1/5

Write 60 out of 105 as a percent, rounded to the nearest tenth.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q2/5

27 is what percent of 60?

A. 45%

B. 60%

C. 33%

D. 27%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q3/5

True or false?

55 out of 200 is 27.5%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q4/5

True or false?

40 out of 95 is 40%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSS-ID.B.5

FORWARD 1

Use a two-way table to find conditional probabilities and test independence • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A student from the table is chosen at random.

Find P(plays a sport | Grade 10).

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.4 • Set 62 • FORWARD 1 • Q1/5

Are "Grade 10" and "plays a sport" independent? Explain with probabilities.

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.4 • Set 62 • FORWARD 1 • Q2/5

What is P(Grade 9 | plays a sport)?

	Plays a sport	Does not	Total
Grade 9	45	55	100
Grade 10	60	40	100
Total	105	95	200

A. About 0.43

B. 0.225

C. 0.525

D. 0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.4 • Set 62 • FORWARD 1 • Q3/5

True or false?

If P(B) > 0, events A and B are independent exactly when P(A | B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.4 • Set 62 • FORWARD 1 • Q4/5

True or false?

P(A | B) always equals P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.4 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSS-ID.B.5

FORWARD 2

Explain conditional probability and independence in everyday language • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

Explain in everyday language what P(late | rain) > P(late) means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.5 • Set 62 • FORWARD 2 • Q1/5

A and B are independent, P(A) = 0.3, and P(B) = 0.5.

Find P(A and B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.5 • Set 62 • FORWARD 2 • Q2/5

Which pair of events is most likely independent?

A. Height and shoe size

B. Studying and passing a test

C. Flipping heads and rolling a 6

D. Rain and carrying an umbrella

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.A.5 • Set 62 • FORWARD 2 • Q3/5

True or false?

If A and B are independent, then P(A and B) = P(A) · P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.5 • Set 62 • FORWARD 2 • Q4/5

True or false?

If P(A | B) = 0.8 and P(A) = 0.4, then A and B are independent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.A.5 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSS-ID.B.6.a • INTERPRETING CATEGORICAL & QUANTITATIVE DATA

Fit a linear, quadratic, or exponential function to data and use it to solve problems

MAIN	Algebra I main question	**HSS-ID.B.6.a**
BACKWARD 1	Describe patterns in scatter plots	**8.SP.A.1**
BACKWARD 2	Write linear and exponential functions through given points	**HSF-LE.A.2**
FORWARD 1	Use logarithms with an exponential model fitted to data	**HSF-LE.A.4**
FORWARD 2	Fit a sinusoidal model to periodic data	**HSF-TF.B.5**

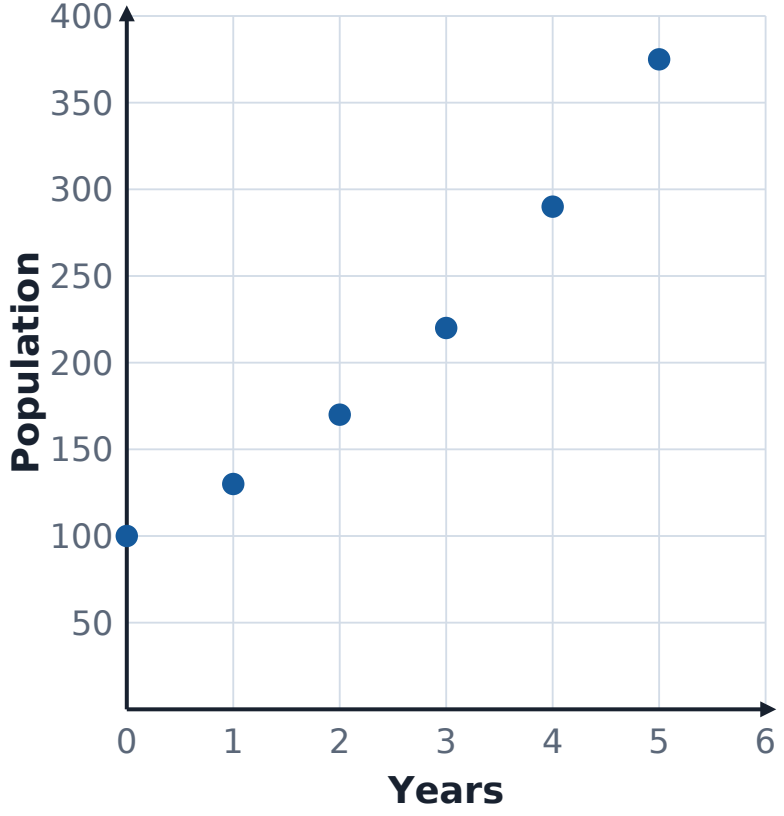
SET 63 • HSS-ID.B.6.a

MAIN

Algebra I main question • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The scatter plot shows a population over 5 years.

Which type of function fits best? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q1/5

An exponential model for the data is $P(t) = 100(1.3)^t$.

Use it to predict the population at $t = 8$, to the nearest whole number.

$P(8) \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q2/5

Data: x = 0, 1, 2, 3, 4 and y = 2, 5, 10, 17, 26.

Which quadratic function fits these data exactly? Write it.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q3/5

A scatter plot rises and then falls in a symmetric curve.

Which model fits best?

A. Quadratic

B. Exponential growth

C. Exponential decay

D. Linear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q4/5

True or false?

Using a model to predict far outside the range of the data is always reliable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 63 • MAIN • Q5/5